

MATHEMATICS GRADE 10: REAL NUMBERS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.1: Real Numbers
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Classification of numbers – Classification of whole numbers as odd, even, prime and composite – Classification of real numbers as rational and irrational – Reciprocal of numbers – Reciprocal of real numbers by division – Reciprocal of real numbers using mathematical tables and calculators – Application of reciprocal of real numbers in mathematical computations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) classify whole numbers as odd, even, prime and composite in different situations, b) classify real numbers as rational and irrational in different situations, c) determine the reciprocal of real numbers by division, d) determine the reciprocal of real numbers by use of mathematical tables and calculators, e) apply reciprocal of real numbers in mathematical computations, f) promote the use of real numbers in day-to-day activities.
Core Competencies	– Communication and Collaboration: the learner discusses with peers to identify and categorise whole numbers as odd, even, prime, composite, rational, and irrational. – Critical Thinking and Problem Solving: the learner uses the reciprocal of real numbers to work out mathematical tasks. – Digital Literacy: the learner works out the reciprocal of real numbers from calculators and other digital devices.
Core Values	– Unity: the learner discusses in a group and categorises real numbers as rational and irrational numbers. – Responsibility: the learner takes care of calculators and mathematical tables when working out the reciprocal of real numbers.
Science & Engineering Practices	– Asking questions and defining problems: students ask why different kinds of numbers are needed for money, sharing and measuring, and why $\sqrt{2}$ never ends. – Developing and using models: students build and revise a number-system map (whole numbers, rational, irrational, real) and a reciprocal model. – Analyzing and interpreting data: students examine decimal expansions of fractions and roots from calculators to spot

	terminating, recurring and non-terminating patterns. – Using mathematics and computational thinking: students use division, tables and calculators to find reciprocals and solve price, speed and rate problems. – Constructing explanations: students justify classifications of numbers using evidence and reasoning. – Engaging in argument from evidence: students defend whether a number is rational or irrational in group discussions.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (self-esteem): the learner participates in the discussion on the concepts of rational and irrational numbers.
Career Connections	– Shopkeeper/Entrepreneur (duka, market traders in Gikomba or Kisumu): uses unit prices, change and percentages daily. – Banker/Sacco officer: calculates interest and percentages on savings and loans. – Accountant and Actuary: classifies and computes with real numbers in finance and risk. – Engineer and Surveyor: uses reciprocals and irrational numbers such as $\sqrt{2}$ in measurement and design. – Chef/Caterer and Pilot or Matatu operator: uses fractions, rates, and speed-distance-time calculations.
Focus for Lessons	This unit builds understanding of the real number system by starting from everyday Kenyan money and measurement situations such as shopping at a duka and sharing food. Students classify whole numbers and real numbers, discover why some numbers like $\sqrt{2}$ never end, and learn to find and apply reciprocals using division, tables and calculators. Learning follows a predict, observe, explain, DQB and model-building cycle where students keep revising a number-system model.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do we use real numbers in everyday life, and why are reciprocals important? KICD KEY INQUIRY QUESTIONS: 1. How do we use real numbers in day-to-day activities? SUPPORTING QUESTIONS (DQB): 2. What types of numbers exist? 3. How do we classify whole numbers? 4. What's the difference between rational and irrational numbers? 5. Why does $\sqrt{2}$ go on forever? 6. What is a reciprocal? 7. How do reciprocals help us solve problems? 8. Where do we use real numbers in daily life?
Anchoring Phenomenon	A learner is sent to a duka in Kisumu to buy 3 items at Ksh 45 each and pays with a Ksh 200 note. She must work out the total cost ($3 \times 45 = \text{Ksh } 135$) and the change (Ksh 65). The shopkeeper then says, 'the unit price is Ksh 15,' for a pack of 6 mandazi costing Ksh 90, and the learner wonders what 'unit price' really means. Throughout the transaction different kinds of numbers appear: whole numbers for counting items, fractions for half a kilo of sugar, and decimals on the calculator and price tags such as Ksh 22.50. It is puzzling that one simple shopping trip needs so many types of numbers, and that the numbers are not all the same kind. Why do we need whole numbers, fractions and decimals for different situations, and how are they related?
Storyline Thread	Lesson 1: Shopping and money at the duka. Students predict which kinds of numbers appear in a purchase (3 items at Ksh 45, Ksh 200 paid, change, unit price Ksh 15), sort them, and launch the Driving Question Board and initial model of types of numbers. Lesson 2: Classifying whole numbers as odd, even, prime and composite using Kenyan contexts (seating in a matatu, packing eggs, sharing maize). Students test and justify classifications, including the sieve of Eratosthenes, and revise the model.

	Lesson 3: Real numbers as rational and irrational. Students convert fractions to decimals (terminating and recurring) and compare with $\sqrt{2}$ and π, asking why some decimals never end. Lesson 4: Why does $\sqrt{2}$ go on forever? Students use calculators and square-measurement of a field to investigate irrational numbers, argue from evidence, and refine the number-system map of real numbers. Lesson 5: What is a reciprocal? Students find reciprocals by division ($1 \div n$) using sharing situations such as unit price and recipes, observing that a number times its reciprocal is 1. Lesson 6: Finding reciprocals using mathematical tables and calculators ($1/x$ key), comparing methods and accuracy, and caring for the tables and calculators. Lesson 7: Applying reciprocals and real numbers to speed, time, rate and interest problems (Nairobi to Mombasa journey, Sacco savings). Students answer the Driving Question, review the DQB, and finalise the unit model.
Supporting Phenomena	– Dividing pizza equally: How many slices does each person get when a pizza is shared among friends, and what if the slices cannot be shared evenly? – Measuring ingredients: Why do recipes for mandazi use fractions such as $\frac{1}{2}$ cup of sugar or $\frac{3}{4}$ kg of flour? – Calculating speed: Distance $\div$ time = speed, so what if we want time? A matatu from Nairobi to Mombasa shows how the reciprocal helps. – Banking interest: How do we calculate percentages and interest on Sacco or M-Pesa savings? – $\sqrt{2}$ on a calculator: Why does the decimal for $\sqrt{2}$ go on forever without repeating?

ATTRIBUTION AND LICENSING

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LESSON 1 (40 minutes): Shopping at the duka: what kinds of numbers?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Launch the unit by getting learners to notice that one simple shopping trip at a duka in Kisumu uses whole numbers, fractions and decimals, each for a different purpose, and that unit price is found by division. Learners begin the Driving Question Board (DQB) and an initial 'unit model' of types of numbers.
Knowledge	Numbers come in different types: whole numbers (counting items and money, e.g. 3, 45, 135, 200, 65), fractions (parts of a whole, e.g. 1/2 kg of sugar) and decimals (prices such as Ksh 22.50). Unit price is the cost of one item, found by dividing: total cost ÷ number of items ($90 \div 6 = \text{Ksh } 15$). Fractions and decimals can be two ways of writing the same number ($22.50 = 22 \frac{1}{2} = 45/2$).
Skills	Predicting, sorting and classifying numbers by type, calculating with money ($3 \times 45 = 135$; $200 - 135 = 65$; $90 \div 6 = 15$; $1/2 \times 160 = 80$), asking questions, and drawing an initial model. NGSS-aligned practices: Asking Questions and Defining Problems; Developing and Using Models; Analyzing and Interpreting Data; Constructing Explanations.
Attitudes	Communication and Collaboration: the learner discusses with peers to identify and categorise numbers. Unity: the learner works with the group to sort numbers. Life Skills (self-esteem): the learner participates in the discussion and shares ideas confidently, even tentative ones. Curiosity about everyday mathematics.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This is the launch lesson. It presents the anchoring phenomenon (Achieng's trip to the duka near Kibuye Market, Kisumu) and uses only prior primary-school knowledge of whole numbers, fractions and decimals. It opens the DQB with questions 1-8 (as supplied, not renumbered) and starts the initial unit model. Question 3 (How do we classify whole numbers?) leads into Lesson 2. Questions on rational and irrational numbers, reciprocals and uses of real numbers are parked on the DQB for later lessons. The ideas are only introduced here, not taught.
Safety Notes	No hazardous materials. Use paper money cut-outs or price cards instead of real money. Keep walkways clear during group movement. Learners handle calculators and the shared charts with care (Responsibility). Be sensitive to learners from families with limited money: frame the duka scenario as a mathematics puzzle, not a judgement about spending.

B. LESSON OVERVIEW

Ms. Wanjiku opens with a story about Achieng, a Grade 10 learner sent by her mother to a duka near Kibuye Market in Kisumu. Achieng buys 3 items at Ksh 45 each, pays with one Ksh 200 note, and works out the total (Ksh 135) and the change (Ksh 65). Mama Atieno sells 6 mandazi for Ksh 90, and Mzee Odhiambo the shopkeeper says 'the unit price is Ksh 15'. Achieng wonders what 'unit price' really means. She also buys 1/2 kg of sugar (Ksh 160 per kg, so Ksh 80) and sees a tea-leaf sachet price tag of Ksh 22.50. In the Predict Phase learners individually predict which kinds of numbers appear in the purchase. In the Observe Phase, in groups (Lake Victoria Group, Kit Mikayi Group, Impala Group, Rift Valley Group), they work through the transaction data cards, check each calculation and sort every number into whole number, fraction or decimal. In the Explain Phase the class discusses why different number types are needed and what unit price means (a division: $90 \div 6$). Learners then generate questions to launch the Driving Question Board, and finish by drawing an initial unit model of types of numbers. The lesson ends with the puzzle that guides the unit: why do we need whole numbers, fractions and decimals, and how are they related?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners listen to the story of Achieng at the duka near Kibuye Market in Kisumu. The teacher reads the data aloud and writes it on the board: 3 items at Ksh 45 each, a Ksh 200 note, the change, 6 mandazi for Ksh 90 with Mzee Odhiambo saying 'the unit price is Ksh 15', 1/2 kg of sugar, and a tag of Ksh 22.50. Working alone for 3 minutes, each learner writes in their exercise book: (1) which kinds of numbers they think appear in this one shopping trip, (2) one example of each kind, and (3) what they think 'unit price' means. They then share with a partner and record any difference in their predictions. Learners are not told the answers and are not marked on correctness.	Say: 'Today we begin with a puzzle from a duka in Kisumu. Achieng's mother has sent her to buy a few things. Listen for every number you hear.' Read the story slowly and write each number on the board as you say it. Then say: 'Do not look at anyone's page yet. Write down what kinds of numbers you think this trip needs. There are no wrong predictions today.' Give 3 minutes of silent writing. Then say: 'Turn to your partner and compare.' Circulate and listen for words such as 'counting numbers', 'half', 'cents' or 'decimal point'. Cold-call 4 learners from different parts of the room: 'Tell us one kind of number you predicted and why.' After each answer WAIT TIME of 10-15 seconds before asking another learner whether they agree. Record every prediction on the board without judging. Ask: 'What do you think unit price means?' and write 3 ideas on a chart titled 'Our first ideas'.	Prediction with partner talk (Think-Write-Pair-Share). Writing a prediction before any teaching makes learners commit to their prior knowledge from primary school (whole numbers, fractions, decimals) so they can compare it with the evidence later. Hearing a partner's different prediction creates productive disagreement that motivates the investigation. Recording predictions publicly shows that all ideas are valued (Life Skills: self-esteem).	Observe whether learners (a) name at least two types of number, (b) give a correct example from the story for each type (e.g. 3 or 45 as counting numbers, 1/2 as a fraction, 22.50 as a decimal), and (c) attempt a meaning for 'unit price'. Note on a class list which learners predict only whole numbers, as this shows who may need support to see fractions and decimals in the data. Collect the 'Our first ideas' chart for reuse in the Explain Phase and Model Building Phase.
Observe Phase  VIDEO: Classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar	In groups of 4-5 (Lake Victoria Group, Kit Mikayi Group, Impala Group, Rift Valley Group), learners receive a set of 'duka data cards': Card 1: 3 items at Ksh 45 each, Ksh 200 note. Card 2: 6 mandazi cost Ksh 90, unit price Ksh 15. Card 3: sugar Ksh 160 per 1 kg, 1/2 kg bought. Card 4: tea-leaf sachet tag Ksh 22.50, 2 sachets = Ksh 45. Learners (1) calculate and check each step: total $3 \times 45 =$ Ksh 135; change $200 - 135 =$ Ksh 65; unit price $90 \div 6 =$ Ksh 15; half a kilo of sugar $1/2 \times 160 =$ Ksh 80; $2 \times 22.50 =$ Ksh 45. (2) Cut out	Hand out data cards and number strips. Say: 'You are Achieng's maths team. First check the calculations on each card, then sort every number into a column. Be ready to defend where you put each number.' Move to each group. Ask: 'Which number did you place in the fraction column, and what is it measuring?' and 'What did you do to the 90 and the 6 to get 15?' If a group puts 22.50 under whole numbers, ask: 'Can you count Ksh 22.50 in whole shillings only? What does the .50 tell us?' Do not give the answer.	Card sort with data tables (Analyzing and Interpreting Data). Physically sorting numbers into categories makes learners decide what each number type does in the story, rather than memorising names. Checking every calculation first connects the numbers to meaning. Star-marking disagreements turns uncertainty into the next question for the whole class.	Look for: (a) the calculations $3 \times 45 = 135$, $200 - 135 = 65$, $90 \div 6 = 15$, and $1/2 \times 160 = 80$ are correct; (b) 3, 6, 45, 90, 135, 200, 65 and 15 are placed under whole numbers; (c) 1/2 is placed as a fraction and 22.50 as a decimal; (d) learners give a purpose for each type, such as 'counting', 'part of a whole' or 'price with cents'; (e) at least one group links 22.50 to 22 1/2 or 45/2. Record which groups needed prompting about the decimal.

readings	each number that appears (3, 45, 135, 200, 65, 6, 90, 15, 1/2, 160, 22.50) and sort them on a mat into three columns labelled 'whole number', 'fraction' and 'decimal'. (3) Write beside each number what it is doing in the story (counting items, naming money, a part of a kilo, a price with cents). Groups notice that 22.50 can also be written 22 1/2 or 45/2, and that 'unit price' needed a division.	Use WAIT TIME of 10-15 seconds after each probing question before offering a hint. Prompt: 'Is there another way to write 22.50?' and lead them to 22 1/2 or 45/2 only if they are stuck, linking to 2 sachets = Ksh 45. After 10 minutes, cold-call 3 groups to report where they placed one number and why. Note any numbers groups disagree about (for example 22.50 or 1/2) and mark them with a star on the board for the Explain Phase.		
Explain Phase  VIDEO: Converting Fractions into Decimals <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Converting fractions to decimals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class comes together to make sense of the sorting. Learners compare their sorted mats in a gallery walk of 3 minutes, using sticky notes to mark one thing that matches and one that differs from their own group. Then in a whole-class discussion they explain: (1) why Achieng needs whole numbers for counting items and money, fractions for part of a kilo, and decimals for prices such as Ksh 22.50; (2) how fractions and decimals are related (1/2 and 0.5; 22.50 and 22 1/2); (3) what unit price means: the cost of one item, found by dividing total cost by number of items, $90 \div 6 = \text{Ksh } 15$. Each learner then writes a 2-3 sentence explanation in their book that begins: 'Achieng's trip needs different kinds of numbers because ...' and revisits their prediction, marking what was right and what changed. Learners also note an unresolved puzzle: are all numbers like these, or are there other kinds?	Say: 'Walk around and look at other groups' mats. Place a green sticky note where you agree and a yellow one where you see something different.' After 3 minutes bring the class together. Ask: 'Why could Achieng not use only whole numbers for this trip?' Cold-call 3 learners, each time with WAIT TIME of 10-15 seconds, then ask: 'Who can add to that, or disagree?' Point to a starred disagreement, for example 22.50: 'Some groups put 22.50 in the decimal column. What does the .50 stand for?' Guide to 'half a shilling, so 22 and a half'. Write $1/2 = 0.5$ and $22.50 = 22 \frac{1}{2}$ on the board. Ask: 'Mzee Odhiambo said the unit price is Ksh 15. What did he do to Ksh 90 and 6 mandazi?' Cold-call 2 learners. Write 'unit price = total cost $\div$ number of items' and show $90 \div 6 = 15$. Do not explain reciprocals; say: 'There may be another way to write this division. We will return to it later.' Then give 4 minutes for the written explanation. Say: 'Look back at what you predicted. Circle what you got right and underline what you would change.' Finish with the unit puzzle: 'Why do we need whole numbers, fractions and	Claim-evidence-reasoning through whole-class discussion (Constructing Explanations). Learners state a claim (different numbers are needed), support it with evidence from the duka data (3 items, 1/2 kg, Ksh 22.50), and reason about purpose (counting, part of a whole, price with cents). Comparing the written explanation with the first prediction shows learners how their thinking has changed, which builds confidence (self-esteem).	Check written explanations for: (a) at least two number types named with correct examples from the story; (b) a purpose given for each type; (c) unit price described as dividing the total cost by the number of items, with $90 \div 6 = 15$; (d) recognition that 1/2 and 0.5, or 22.50 and 22 1/2, can mean the same quantity. Sort responses into three piles: complete, partly complete, needs support. Use the piles to group learners in Lesson 2 and to decide who needs a quick revisit of fractions and decimals.

		decimals, and how are they related?		
Driving Question Board (DQB) Creation  VIDEO: Converting percents to decimals & fractions example <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Converting Fractions into Decimals Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the story and the puzzle and write 2 questions each on sticky notes, beginning with the words: 'I wonder ...' or 'Why ...?' They share them in their groups, pick the group's best 2 questions, and place them on the DQB chart headed 'How do we use real numbers in everyday life, and why are reciprocals important?'. Learners cluster similar notes. The teacher then introduces the prepared starter questions and learners match their own to them: 2. What types of numbers exist? 3. How do we classify whole numbers? 4. What's the difference between rational and irrational numbers? 5. Why does $\sqrt{2}$ go on forever? 6. What is a reciprocal? 7. How do reciprocals help us solve problems? 8. Where do we use real numbers in daily life? Learners vote with a dot for the question they most want answered.	Say: 'Good mathematicians ask questions. What are you still wondering about Achieng's trip? Write 2 questions on sticky notes. Start with I wonder or Why.' Give 3 minutes and then WAIT TIME of 10-15 seconds before asking groups to choose. Cold-call 4 groups: 'Read out one question and tell us what in the story made you ask it.' Place notes on the DQB under the main driving question and key inquiry question 1: How do we use real numbers in day-to-day activities? Then add the numbered questions 2-8 on the chart as they are supplied without renumbering. Link them to learners' notes: 'Sarah asked about numbers like half a kilo, and that connects to question 2, What types of numbers exist?' Point out that question 3 will be answered next lesson. Say: 'We will come back to this board each lesson and move questions across when we have evidence.' Give each learner one dot to vote. Count aloud the top 2 questions.	Driving Question Board (Asking Questions and Defining Problems). Learners' own questions are placed beside the teacher's questions so that the unit is driven by what they want to know. The DQB acts as a living record of the learning path and is revisited at the end of each lesson, helping learners see how their understanding grows.	Check each learner's sticky notes: do they ask a question linked to the data (types of numbers, unit price, fractions versus decimals)? Look for questions that go beyond recall, such as 'Why do we need fractions if we have decimals?' Note which of questions 2-8 attract the most votes, and which learners did not contribute, so you can invite them in the next lesson. Photograph or keep the DQB chart.
Model Building Phase  VIDEO: Dividing a decimal by a whole number with fraction models <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES	Individually, learners draw an initial unit model of types of numbers on a half-page of paper, using the duka evidence. The model has a main box 'Numbers at the duka' with branches to 'whole numbers (3, 45, 135, 200, 65, 15)', 'fractions (1/2 kg)' and 'decimals (Ksh 22.50)'. Under each branch they write what the number does in Achieng's trip, and add a labelled arrow linking 'fraction' and 'decimal' ($1/2 = 0.5$; $22.50 = 22 \frac{1}{2}$) and a note 'unit price = $90 \div 6 = \text{Ksh } 15$'. They add a question	Say: 'You have evidence from Achieng's trip. Now draw a model of the types of numbers it needed. Your model should show the types, an example of each, and what each one is for.' Draw the main box and three branches on the board as a prompt only, leaving the content blank. Walk around and ask: 'Where on your model do you show that half a kilo can also be written as 0.5?' and 'Where does unit price fit in?' WAIT TIME of 10-15 seconds before prompting. Cold-call 3 learners to	Developing and Using Models (initial unit model). Drawing the relationship between number types makes learners' current thinking visible and gives a base to revise. Leaving a 'other kinds of numbers?' box signals that the model is incomplete, which encourages learners to look for gaps and motivates later lessons.	Check each model for: (a) three types of number, each with a correct example from the duka data; (b) a purpose written for each type; (c) a link showing that fractions and decimals can name the same quantity; (d) unit price shown as a division ($90 \div 6 = 15$); (e) a question or gap marked. Use a quick tally of learners with 4-5 features, 2-3 features, or fewer. Use the tally to plan support in Lesson 2.

for similar videos  EXERCISE: Write common decimals as fractions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	mark box for 'other kinds of numbers?' Learners then pair up, explain their model to a partner in 1 minute each, and add one improvement. Models are kept in the exercise book or on the class wall as the unit model, to be revised in each lesson.	explain their model in one sentence. Say: 'This is only our first model. As we learn about odd, even, prime numbers, and numbers that go on forever, we will change it.' Return to the DQB and ask: 'Which question might we answer first?' and guide them to question 3. Close by re-reading the puzzle and ask them to think about the question: 'Can you sort the whole numbers in the story further?'		
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D. TEACHER REFLECTION

1) Which learners predicted only whole numbers, and did the card sort help them see fractions and decimals? 2) Did learners explain unit price as a division, or did they simply remember 15? 3) Were the starred disagreements (such as 22.50) resolved through evidence rather than by the teacher telling? 4) Which DQB questions did learners add that I did not expect, and how will I use them in later lessons? 5) Did the Kenyan context (Kibuye Market, mandazi, sugar) make the mathematics accessible to all, including learners who rarely buy from the duka? 6) What would I change about group sizes, wait time or materials next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed in Achieng's trip to the duka: she bought 3 items at Ksh 45 each (total Ksh 135), paid Ksh 200 and got Ksh 65 change; 6 mandazi cost Ksh 90 so the unit price is Ksh 15; 1/2 kg of sugar cost Ksh 80; a tea-leaf sachet tag was Ksh 22.50. Whole numbers, a fraction and a decimal all appeared.
What did I learn?	What we learned: numbers come in different types, each with its own purpose: whole numbers for counting items and money, fractions for parts of a whole, and decimals for prices with cents. Fractions and decimals can name the same quantity ($1/2 = 0.5$; $22.50 = 22 \frac{1}{2}$). Unit price is found by dividing the total cost by the number of items.
How does this explain the phenomenon?	Our explanation so far: Achieng's trip needs different kinds of numbers because counting, sharing a whole into parts and pricing with cents are different jobs. Unit price is a division, $90 \div 6 = \text{Ksh } 15$. We still wonder how whole numbers can be classified, what other kinds of numbers exist, and why reciprocals matter. These are on our DQB.

LESSON 2 (40 minutes): Classifying Whole Numbers: Odd, Even, Prime and Composite

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners classify whole numbers as odd, even, prime or composite by counting factors, use the sieve of Eratosthenes to find the primes from 1 to 50, and revise the unit model of number types using evidence from Achieng's duka trip.
Knowledge	a) classify whole numbers as odd, even, prime and composite in different situations. Even numbers are divisible by 2 and odd numbers are not. A prime has exactly two factors (1 and itself) and a composite has more than two factors. 1 is neither prime nor composite, and 2 is the only even prime.
Skills	Communication and Collaboration: the learner discusses with peers to identify and categorise whole numbers as odd, even, prime, composite. Critical Thinking and Problem Solving: listing factors, testing claims with examples and counterexamples, and justifying a classification with evidence. Also constructing explanations and using patterns (NGSS SEPs).
Attitudes	Unity: the learner discusses in a group and categorises numbers together, using group names Lake Victoria, Kit Mikayi, Impala and Rift Valley. Life Skills (self-esteem): the learner participates in the discussion and shares reasoning even when unsure.
Key Inquiry Question	How do we use real numbers in day-to-day activities? (Supporting DQB question 3: How do we classify whole numbers?)
Purpose in Storyline	Lesson 1 sorted the whole numbers from Achieng's duka transaction and launched DQB question 3. This lesson answers that question: it gives a rule for classifying whole numbers by counting factors. It prepares Lesson 3, where whole numbers are seen as special rational numbers ($n = n/1$), and it adds evidence for the final answer to the Driving Question.
Safety Notes	Low-risk lesson. Learners use pencils, paper, 17 counters (bottle tops or maize grains) and charts. Keep walkways clear when learners move to build the gallery walk. Learners take care of the shared charts and counters (Responsibility).

B. LESSON OVERVIEW

Achieng's duka trip used many whole numbers: 3, 6, 15, 45, 65, 90, 135 and 200. Learners predict how these numbers might be sorted, then gather evidence by listing factors, testing real Kenyan situations (a 14-seater matatu, a tray of 30 eggs, 17 cobs of maize shared among groups) and building the sieve of Eratosthenes for 1 to 50. In the Explain phase learners write a rule using factor counts and justify borderline cases (1 and 2). They then add evidence to the DQB and revise the unit model to show how whole numbers are classified.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Finding factors and multiples Source: Khan Academy (English)	Learners recall the duka transaction from Lesson 1, where the whole numbers 3, 6, 15, 45, 65, 90, 135 and 200 were sorted. Each learner silently writes a prediction for the question: 'Which	Say: 'Remember Achieng at Mzee Odhiambo's duka near Kibuye Market. She counted 3 items, paid Ksh 200 and got Ksh 65 change. Today we ask: are all these whole numbers the same kind?' Write 3,	Eliciting prior ideas with Predict-then-Justify: committing to a prediction in writing before instruction exposes learners' existing ideas about odd, even and prime (for example, 'prime means	Observable indicators: (1) at least 6 of 8 numbers placed in a pile with a stated reason on the slip; (2) learners use the words 'divide', 'share equally' or 'factor' in their reasons; (3) note which groups

- US curriculum) Search ARES for similar videos HTML: Divisibility Rules to Find Factors (Flexbook) Source: CK-12 Search ARES for similar readings	of these numbers can be shared equally into groups, and which cannot be shared except into 1 group or groups of 1?' They sort the 8 numbers on a slip into two piles of their own choosing and write one reason. Then they discuss in groups (Lake Victoria, Kit Mikayi, Impala, Rift Valley) and agree on one prediction to put on a group chart.	6, 15, 45, 65, 90, 135, 200 on the board. Ask: 'How might you sort these numbers? Write your idea silently.' WAIT TIME 15 seconds, then allow 2 minutes of silent writing. Cold-call 3 learners for their sorting rule, without saying right or wrong: 'Thank you. What made you put 3 apart from 6?' Record every rule on the board. Circulate and note groups using words such as 'odd', 'even', 'divides' or 'only 1 and itself'.	odd'), so these can be tested later with evidence rather than corrected by telling.	sorted only by odd/even and which noticed 'only shares by 1 and itself' for 3. Record 3 common misconceptions on the board for the DQB.
Observe Phase VIDEO: Finding factors and multiples Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Divisibility Rules to Find Factors Practice Source: CK-12 Search ARES for similar readings	Learners gather evidence in three stations (about 5 minutes each), rotating as a group. Station A, Factor Listing: list all factors of 3, 6, 15, 45, 65, 90, 135 and 200 using division (for example, 15: 1, 3, 5, 15; 45: 1, 3, 5, 9, 15, 45; 65: 1, 5, 13, 65) and count them. Station B, Kenyan Situations: (i) the 14-seater matatu has 2 seats per row and 7 rows ($14 = 2 \times 7$), so can the passengers be seated in equal rows of other sizes? (ii) a tray of 30 eggs: find all ways to pack equal rows ($30 = 2 \times 3 \times 5$); (iii) use 17 counters as 17 cobs of maize and try to share them equally among 2, 3, 4 up to 16 people, recording the remainder each time. Station C, Sieve of Eratosthenes: on a 1 to 50 grid, cross out 1, then circle 2 and cross out its multiples, circle 3 and cross out its multiples, then do the same for 5 and 7. The numbers left circled or uncrossed are the primes. They record a table: number, factors, number of factors, odd or even, prime or composite.	Set up stations before class and give each group a recording table. Say: 'You are collectors of evidence today. Do not just tell me a number is prime; show me its factors.' At Station A, ask: 'How do you know you have found all the factors of 45? How can you check pairs such as 1×45, 3×15, 5×9?' At Station B, ask: 'What happened when you tried to share 17 cobs of maize into groups of 2 to 16? What remains each time?' WAIT TIME 10 seconds after each question. At Station C, ask: 'Why do we stop crossing out after 7 for numbers up to 50?' (Hint: the next prime, 11, has $11 \times 11 = 121$, which is more than 50.) Cold-call 2 groups per station to share one finding. Watch for the mistakes 'all odd numbers are prime' (15, 45, 65, 135 are odd composites) and 'prime numbers are always odd' (2 is even and prime). Do not correct yet; ask the group to test with their factor table.	Evidence gathering with a Data Table and Pattern Finding: completing the same table (factors, count, type) for each number lets learners notice that primes have exactly 2 factors, composites have more than 2, and that oddness or evenness does not decide primality. The sieve gives a visual, pattern-based check from a different angle.	Observable indicators: (1) factor lists for 15, 45 and 65 are complete and correct; (2) the 17-cob table shows a remainder for every group size from 2 to 16 and learners say '17 has only 1 and 17'; (3) the sieve grid shows exactly 15 primes between 1 and 50 (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47) with 1 crossed out and not counted; (4) learners connect the 14-seater matatu ($14 = 2 \times 7$) and 30 eggs ($30 = 2 \times 3 \times 5$) to even composites. Tick the checklist for each group.
Explain Phase VIDEO: Least common	Groups use their tables and sieve to write a classification rule in their own words and complete the	Say: 'Use your evidence, not your memory. State a claim, give the evidence from your table, and give	Claim-Evidence-Reasoning with Counterexample Testing: learners must support a classification with	Observable indicators: (1) rule sentences state 'prime = exactly two factors' and 'composite = more

multiple: repeating factors <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Prime and Composite Numbers (Flexbook) Source: CK-12 Search ARES for similar readings	sentence frames: 'A number is even when _____. A number is odd when _____. A number is prime when _____. A number is composite when _____. Then they classify all 8 duka numbers: 3 is prime and odd; 6 is even composite; 15 is odd composite (3×5); 45 is odd composite ($3 \times 3 \times 5$); 65 is odd composite (5×13); 90 is even composite; 135 is odd composite; 200 is even composite. Groups then argue two puzzles: 'Is 1 prime?' (1 has only one factor, so it is neither prime nor composite) and 'Is 2 prime, even, or both?' (2 has exactly two factors and is even, so it is the only even prime). Each group presents a claim, evidence and reasoning (CER) for one number and explains what the 17 cobs, the 14-seater matatu and the 30 eggs show.	your reasoning.' Provide the CER frame on the board: 'I claim _____ is _____. My evidence is the factors _____. This shows _____ because _____. Run an Argue-From-Evidence round: ask 4 groups (cold-call) to present. Ask: 'Someone said all odd numbers are prime. Who has a counterexample?' WAIT TIME 15 seconds; expect 15, 45, 65 or 135. Ask: 'Someone said all prime numbers are odd. What does 2 say?' WAIT TIME 10 seconds. Ask: 'If Achieng had 17 cobs of maize, why would sharing equally among friends be impossible, but 12 eggs shared by 3 gives 4 each?' Press for the words 'factor' and 'exactly two factors'. Confirm the correct definitions only after learners have given them, and write the final rule on the board. Link back: 'Achieng's numbers, 15 Ksh unit price and Ksh 65 change, are both odd composites.'	factor evidence and then try to break each generalisation with a counterexample. This builds a precise, evidence-based definition and corrects the misconceptions from the Predict phase.	than two factors', not 'odd numbers'; (2) learners give at least one counterexample to each false claim; (3) 1 is correctly stated as neither prime nor composite and 2 as the only even prime; (4) all 8 duka numbers correctly classified. Exit check: on a slip, classify 27, 29 and 100 with factors (27 odd composite, 29 odd prime, 100 even composite) - aim for 80% correct.
Driving Question Board (DQB) Creation  VIDEO: Patterns with multiplying even and odd numbers <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Prime and Composite Numbers Practice Source: CK-12	Learners return to the DQB started in Lesson 1. Groups place sticky notes under question 3, 'How do we classify whole numbers?', with the answer they found and the evidence (for example, 'Prime = exactly 2 factors: 17 cobs, 29 on the sieve'). They also add new questions that arise, such as 'Are there infinitely many primes?', 'Why is 2 the only even prime?', or 'Can fractions be classified in the same way?' They mark which questions are now answered (3) and which are partly answered (2: What types of numbers exist?; 8: Where do we use real numbers in daily life?), and which are for later lessons (4, 5, 6, 7).	Say: 'Look at our DQB. Which question did we answer today, and what is your proof?' Point to question 3 and wait: WAIT TIME 10 seconds. Cold-call 3 learners to read an answer card and its evidence. Ask: 'What new question has today left you with?' Collect 4 to 5 new questions and read them aloud, grouping them. Tell learners: 'Notice that question 4, rational and irrational numbers, comes next. Fractions such as $\frac{1}{2}$ kg of sugar and decimals such as Ksh 22.50 have not been classified yet.' Do not answer new questions; place them on the board for later lessons.	Questioning with a Driving Question Board: tracking which questions are answered with evidence and which remain open keeps learners' curiosity linked to the anchoring duka phenomenon and shows the unit as one connected storyline rather than separate topics.	Observable indicators: (1) at least one evidence-based answer card attached to question 3 per group; (2) learners write at least one new, genuine question (not a restatement); (3) learners correctly tell which DQB questions are answered, partly answered or still open. Check cards for the misconception 'odd means prime'.

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Model Building Phase VIDEO: Intro to factors & divisibility <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Prime and Composite Numbers (PLIX) Source: CK-12 Search ARES for similar readings	Learners revise the unit model of number types begun in Lesson 1, in which whole numbers, fractions and decimals appeared in Achieng's duka transaction. In groups they extend the whole-number branch on the chart into a classification tree: whole numbers split into even and odd; also into prime (2, 3, 5, 7, 11 ... 47), composite (6, 15, 45, 65, 90, 135, 200) and the special number 1 (neither). They place the 8 duka numbers, 14 seats, 30 eggs and 17 cobs on the tree and write one sentence on how factor counting decides each place. Finally they write a 'Still to figure out' note: how fractions ($\frac{1}{2}$) and decimals (22.50) fit with whole numbers.	Say: 'Our unit model has a whole-number branch. Today we add detail to it. Show how Achieng's numbers 3, 6, 15, 45, 65, 90, 135 and 200 fit.' Provide chart paper and markers. Circulate and ask: 'Where does 15 go if it is odd? Can a number be both odd and composite? Why does 1 sit apart from the others?' WAIT TIME 10 seconds. Cold-call 2 groups to explain their tree, then ask the class: 'What changed in our model since Lesson 1?' Collect 3 revisions. Close: 'Achieng saw many number types at the duka. Today we sorted the whole numbers; next time we ask how fractions and decimals are related.' Collect charts for display.	Model Revision (Develop and Use Models): extending the unit model with a classification tree shows learners that models are revised as evidence grows, and links the new classification rule back to the original duka phenomenon.	Observable indicators: (1) the model shows both the odd/even split and the prime/composite split, with 1 placed as neither; (2) all 8 duka numbers are correctly placed, with 3 as prime and odd and 6 as even composite; (3) the model includes a sentence stating that classification depends on counting factors; (4) the 'Still to figure out' note refers to fractions or decimals. Use a 3-point rubric (placement, justification, link to the duka).

D. TEACHER REFLECTION

Which learners still believe that odd numbers are prime or that 1 is prime, and which counterexamples (15, 45, 65, 2, 1) changed their thinking? Did the 17 cobs of maize and the sieve make the idea of exactly two factors clear? Which groups needed more time at the factor-listing station? Note the learners who helped others (Unity) and any who were reluctant to share (self-esteem). What will you adjust before Lesson 3, where rational numbers and decimals are introduced?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We listed the factors of the duka numbers 3, 6, 15, 45, 65, 90, 135 and 200. We tried to share 17 cobs of maize equally and always had a remainder. We saw that 14 seats (2×7) and 30 eggs ($2 \times 3 \times 5$) can be arranged in several equal ways. We built the sieve of Eratosthenes for 1 to 50 and found 15 primes.
What did I learn?	Even numbers are divisible by 2 and odd numbers are not. A prime has exactly two factors (1 and itself) and a composite has more than two factors. 1 is neither prime nor composite, and 2 is the only even prime. Being odd does not make a number prime (15, 45, 65 and 135 are odd composites).
How does this explain the phenomenon?	Achieng's duka numbers are classified by counting factors: 3 is prime and odd; 6, 90 and 200 are even composites; 15, 45, 65 and 135 are odd composites. This answers DQB question 3 and adds a classification branch to our unit model of number types, showing that the whole numbers at the duka are not all the same kind.

LESSON 3 (40 minutes): Rational and Irrational Numbers: Why Do Some Decimals Never End?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use long division to convert fractions from the duka ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{3}$, $\frac{1}{6}$, $\frac{1}{7}$, $\frac{1}{15}$, $\frac{1}{45}$) into decimals and compare them with $\sqrt{2}$ and π , so they can classify real numbers as rational or irrational and explain why fractions and decimals are two ways of writing the same rational numbers.
Knowledge	By the end of the lesson the learner should be able to: (b) classify real numbers as rational and irrational in different situations. Rational numbers can be written as $\frac{a}{b}$ (a, b integers, $b \neq 0$) and their decimals either terminate ($\frac{1}{2} = 0.5$, $\frac{1}{4} = 0.25$, $\frac{1}{8} = 0.125$) or recur ($\frac{1}{3} = 0.333...$, $\frac{1}{6} = 0.1666...$, $\frac{1}{7} = 0.142857142857...$, $\frac{1}{15} = 0.0666...$, $\frac{1}{45} = 0.02222...$). Whole numbers are rational because $n = \frac{n}{1}$, and Ksh 22.50 = 22 $\frac{1}{2}$ = $\frac{45}{2}$. Irrational numbers such as $\sqrt{2} = 1.41421356...$ and $\pi = 3.14159265...$ have decimals that never end and never repeat. Rational and irrational numbers together make up the real numbers.
Skills	Performing long division to convert fractions to decimals; spotting repeating blocks; writing recurring decimals with a dot or bar; classifying numbers as rational or irrational using evidence; Developing and using models; Analyzing and interpreting data; Constructing explanations; Engaging in argument from evidence. Core competencies: Communication and Collaboration (discussing with peers to categorise numbers as rational and irrational); Critical Thinking and Problem Solving.
Attitudes	Unity: the learner discusses in a group and categorises real numbers as rational and irrational numbers. Life Skills (self-esteem): the learner participates in the discussion on the concepts of rational and irrational numbers. Curiosity about patterns in numbers.
Key Inquiry Question	How do we use real numbers in day-to-day activities? Supporting DQB question 4: What's the difference between rational and irrational numbers? (Also opens DQB question 5: Why does $\sqrt{2}$ go on forever?)
Purpose in Storyline	Lesson 1 showed that Achieng's duka trip uses whole numbers, fractions ($\frac{1}{2}$ kg of sugar) and decimals (Ksh 22.50). Lesson 2 classified whole numbers. In this lesson learners find out how fractions and decimals are related (same rational numbers, written two ways), and meet numbers whose decimals never end or repeat. The puzzle of $\sqrt{2}$ and π is left open for Lesson 4, where learners gather evidence on why $\sqrt{2}$ goes on forever.
Safety Notes	No hazards. Learners work at desks in groups of 4 (Lake Victoria Group, Kit Mikayi Group, Impala Group, Rift Valley Group). Calculators are used with care and returned after the lesson (Responsibility). Remind learners not to round during long division, so the repeating pattern can be seen.

B. LESSON OVERVIEW

Back at the duka in Kisumu, Mzee Odhiambo weighs $\frac{1}{2}$ kg of sugar and the tea sachet is tagged Ksh 22.50. Achieng's calculator shows 0.5 for $\frac{1}{2}$, yet a friend's calculator shows 0.3333333 for $\frac{1}{3}$ and seems never to stop. Learners predict which fractions have decimals that end, then use long division to convert $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{3}$, $\frac{1}{6}$, $\frac{1}{7}$, $\frac{1}{15}$ and $\frac{1}{45}$ into decimals, sorting them into terminating and recurring. They then compare these with $\sqrt{2} = 1.41421356...$ and $\pi = 3.14159265...$, looking at the digits for any repeating block. They conclude that fractions and decimals are two ways of writing the same rational numbers (even $22.50 = \frac{45}{2}$ and $3 = \frac{3}{1}$), while $\sqrt{2}$ and π cannot be written as $\frac{a}{b}$ because their decimals never end or repeat. Learners add new questions to the Driving Question Board, notably DQB 4 and DQB 5, and revise the unit model by splitting real numbers into rational and irrational.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to rational & irrational numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers: rational & irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the duka scene: 1/2 kg of sugar at Ksh 160 per kg, and the Ksh 22.50 tea sachet ($22.50 = 45/2$). Individually, on a sticky note or in their exercise books, they predict: (a) What decimal will 1/2, 1/4 and 1/3 give if we divide the top number by the bottom? (b) Will every fraction's decimal end? (c) Do you think $\sqrt{2}$ and π behave like fractions when written as decimals? They draw a quick prediction line: 'ends' or 'goes on'. Then they share with a partner and record a group prediction in a Lake Victoria, Kit Mikayi, Impala or Rift Valley Group list.	Display on the board: 'Achieng's calculator shows 0.5 for 1/2 but a long string of 3s for 1/3.' Ask: 'Which of these fractions do you think will give a decimal that stops, and which will go on? Write your prediction first.' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking, 'Why might a decimal never stop?' Cold-call 4 learners from different groups and record their predictions on the board under 'Ends' and 'Goes on' without saying who is right. Say: 'Hold on to these predictions; today's evidence will test them.' Circulate and note learners who think all decimals end.	Prediction with reasoning (Predict-Observe-Explain): writing a prediction before the evidence makes learners' prior ideas about fractions and decimals visible, so the later division results create a real need to revise them. Linking back to the duka keeps the numbers meaningful.	Observe whether learners (1) link fractions and decimals as two forms of the same quantity (e.g. $1/2 = 0.5$, Ksh 22.50 = 45/2); (2) give a reason for 'ends' or 'goes on' rather than a guess; (3) show the common misconception that all decimals end or that a decimal must have a pattern. Tally predictions on the board: count how many learners predicted 1/3 would end.
Observe Phase  VIDEO: Classifying numbers: rational & irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners use long division (no calculator at first) to convert $1 \div 2$, $1 \div 4$, $1 \div 8$, $1 \div 3$, $1 \div 6$, $1 \div 7$, $1 \div 15$ and $1 \div 45$. Each group member takes two fractions and then shares their work on a group data table with columns: Fraction Decimal Does it end? Repeating digits. They record: $1/2 = 0.5$, $1/4 = 0.25$, $1/8 = 0.125$ (all stop, remainder 0); $1/3 = 0.333...$, $1/6 = 0.1666...$, $1/7 = 0.142857142857...$ (block 142857, length 6), $1/15 = 0.0666...$, $1/45 = 0.02222...$ (the remainder repeats so the digits repeat). They mark the repeating digits with a dot or bar. Next, they check with a calculator ($1/x$ key or division) and notice the display is rounded. Finally, they look at $\sqrt{2} = 1.41421356...$ and $\pi = 3.14159265...$ printed on the data card, and search for a repeating	Model the first division, $1 \div 8$, on the board step by step, then say: 'Now do your own divisions; watch the remainder at each step. What happens when the remainder becomes 0? What happens when a remainder comes back again?' Circulate and ask each group: 'Which fractions stopped? How do you know?' Prompt: 'In 1/7, which remainder came back? What does that do to the digits?' Call the class together and WAIT TIME of 10-15 seconds before asking: 'Compare $\sqrt{2}$ and π with 1/7. Is there a repeating block?' Cold-call 3 learners to read out one row of their table. Remind groups to take care of calculators (Responsibility). Support learners who stop at 0.3 by asking 'What is the remainder now? Is it 0?'	Data table and pattern-finding: sorting decimals into 'ends' and 'repeats' and linking repeating digits to repeating remainders helps learners notice the structure that makes a number rational. Contrasting these with $\sqrt{2}$ and π , which show no repeating block, builds the idea that there is another kind of number.	Check group tables for correct decimals ($1/8 = 0.125$, $1/7 = 0.142857...$, $1/45 = 0.02222...$). Look for: correct use of dot or bar notation; learners who say a decimal ends because the remainder reaches 0; learners who identify the 142857 block. Ask one learner per group: 'Why did 1/3 never reach remainder 0?' Mark learners who misread 1/6 as 0.16 as needing support with the repeating 6.

	block, finding none in the digits given. They also note whole numbers like 3 and 45 can be written $\frac{3}{1}$ and $\frac{45}{1}$, and $22.50 = \frac{45}{2}$.			
Explain Phase  VIDEO: Worked example: classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Writing fractions as repeating decimals Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners discuss in groups and then as a class to explain what they found. They write a claim-evidence-reasoning statement: 'Claim: Fractions give terminating or recurring decimals. Evidence: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ end; $\frac{1}{3}$, $\frac{1}{6}$, $\frac{1}{7}$, $\frac{1}{15}$, $\frac{1}{45}$ repeat. Reasoning: when dividing, either the remainder becomes 0 or a remainder repeats, so the digits repeat.' They define a rational number as one that can be written as $\frac{a}{b}$ with a, b integers and $b \neq 0$ (including whole numbers, $n = \frac{n}{1}$, and $22.50 = \frac{45}{2}$) and an irrational number as one whose decimal never ends and never repeats, such as $\sqrt{2}$ and π . They state that rational and irrational numbers together make up the real numbers. Each group sorts a set of cards (0.5, 3, 22.50, $\frac{1}{3}$, $\frac{1}{7}$, $\sqrt{2}$, π , 0.0666...) into 'rational' and 'irrational' and justifies each card to another group. Finally they link back: 'Why does one shopping trip need so many types of numbers? Because whole numbers, fractions and decimals are all rational numbers written in different ways.'	Ask: 'What did the division tell us about which decimals repeat? What evidence supports your claim?' Use WAIT TIME of 10-15 seconds, then cold-call 4 learners from different groups to give claims, and 2 to give evidence. Probe: 'Is 3 a rational number? Can you write it as a fraction?' ($3 = \frac{3}{1}$). 'Is Ksh 22.50 rational? Show me as $\frac{a}{b}$.' ($\frac{45}{2}$). Introduce the vocabulary: terminating, recurring, rational, irrational. Say: 'We do not yet know WHY $\sqrt{2}$ never ends; we only know that the digits we have show no end and no repeat. That will be our question for the next lesson.' Facilitate the card sort; challenge groups: 'A friend says 3.14 is irrational because π is 3.14159... Is she right?' (3.14 is rational, an approximation of π). Encourage respectful disagreement (Unity) and invite shy learners to share (self-esteem).	Claim-Evidence-Reasoning with a card sort: learners must justify classifications using their division evidence, so they build the definitions of rational and irrational themselves rather than memorising them. Peer explanation and argument from evidence strengthen communication and collaboration.	Listen for: using ' $\frac{a}{b}$ ' to justify rational; correctly stating that terminating AND recurring decimals are rational; recognising that 3.14 and $\frac{22}{7}$ are rational approximations, not π ; classifying 3 and 22.50 as rational. Use a thumbs signal (up/sideways/down) on 'Is 0.0666... rational?'. Collect an exit ticket: 'Classify $\frac{1}{8}$, $\frac{1}{7}$, $\sqrt{2}$ and give a reason for each.' Note learners who think 'big decimals' are irrational.
Driving Question Board (DQB) Creation  VIDEO: Classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES	Learners return to the Driving Question Board started in Lesson 1. In groups they review the supporting questions and decide which have been partly answered. They add evidence to DQB 4: 'What's the difference between rational and irrational numbers?' (rational decimals end or repeat and can be written as $\frac{a}{b}$;	Point to the board: 'Which of our questions did today's division help us answer? Which question is still bothering you?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to place a sticky note and explain its position. Keep the numbering 1-8 unchanged and add new questions beside the original ones. Say: 'DQB 5 stays	Driving Question Board: making questions public and tracking which are answered keeps learning learner-driven, shows progress towards the driving question, and creates purposeful links to the next lesson ($\sqrt{2}$ evidence) and later lessons (reciprocals).	Check that new questions are mathematically meaningful and connect to evidence from today (e.g. about repeating blocks, $\sqrt{2}$, or reciprocals). Count how many groups moved DQB 4 to 'partly answered' with correct evidence. Note learners who only copy others' questions and prompt them with 'What surprised you in the

for similar videos  HTML: Properties of Rational Numbers versus Irrational Numbers (Flexbook) Source: CK-12  Search ARES for similar readings	irrational ones never end or repeat). They mark DQB 2 (What types of numbers exist?) with new evidence (rational and irrational). They put a star on DQB 5: 'Why does $\sqrt{2}$ go on forever?' as the question we cannot yet answer and write new sticky-note questions such as 'Is there a fraction that equals $\sqrt{2}$?', 'How can we check that $\sqrt{2}$ never repeats?', and 'Do reciprocals of whole numbers like 7 give recurring decimals?' (which links to DQB 6). They place notes under the headings 'Answered so far', 'Partly answered' and 'New questions'.	open; in the next lesson we will gather evidence using a square field.' Ask: 'Which new question connects to reciprocals (DQB 6)?' and highlight that $1/7$ is the reciprocal of 7. Praise questions that are testable.		division?'
Model Building Phase  VIDEO: Intro to rational & irrational numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Properties of Rational Numbers versus Irrational Numbers Practice Source: CK-12  Search ARES for similar readings	Learners revise the unit model (the cumulative number-system map) begun in Lesson 1 and extended in Lesson 2. In groups they redraw it so that 'real numbers' branches into 'rational numbers' and 'irrational numbers'. Under rational they place whole numbers (with $n = n/1$), fractions ($1/2$, $1/4$), terminating decimals (0.5, 0.125, Ksh 22.50 = $45/2$) and recurring decimals ($1/3$, $1/7$, $1/45$). Under irrational they place $\sqrt{2}$ and π with the note 'decimal never ends, never repeats'. They connect Achieng's duka items to the model: 3 items and Ksh 135 (whole numbers), $1/2$ kg of sugar (fraction), Ksh 22.50 (decimal), all rational. They add an arrow labelled '? why' pointing to $\sqrt{2}$ for the next lesson. Groups then compare maps in a gallery walk and add a one-sentence note on what changed since Lesson 2.	Say: 'Redraw your model. Where do whole numbers, fractions and decimals fit? Where do $\sqrt{2}$ and π go, and why?' Circulate and ask: 'Why is 22.50 inside the rational branch?' and 'What evidence puts $\sqrt{2}$ in the other branch?' Give WAIT TIME of 10-15 seconds before each probe. Cold-call 3 groups to explain one change to their model. Use the gallery walk: each learner leaves one 'I notice' and one 'I wonder' note. Close by saying: 'Our model says some numbers never end. Next lesson we will test whether any ending decimal can square to exactly 2.'	Model revision (cumulative concept map): adding the rational/irrational split to the existing map shows how new evidence changes understanding and connects the lessons. The gallery walk builds communication, and the '? why' arrow keeps the open question visible.	Check models for: real numbers split into rational and irrational; whole numbers, fractions and decimals placed as rational; $\sqrt{2}$ and π placed as irrational with a reason; explicit link to the duka items. Rubric: 3 = correct placement with evidence for all; 2 = correct placement, missing reason; 1 = misplaces recurring decimals as irrational. Note common errors, such as putting $1/3$ in the irrational branch because it never ends.

D. TEACHER REFLECTION

After the lesson, check: Did learners use long division correctly and see that a repeating remainder gives repeating digits? Which misconception was most common: 'any decimal that goes on is irrational' ($1/3$ recurs, so it is rational) or ' π equals $22/7$ or 3.14 '? Did learners connect the rational family to Achieng's duka numbers (3 , $1/2$, $22.50 = 45/2$)? Did cold-calling reach quiet learners, and was WAIT TIME of 10-15 seconds used? For the next lesson, prepare the square-field squaring checks ($1.414 \times 1.414 = 1.999396$, $1.4142 \times 1.4142 = 1.99996164$, $1.415 \times 1.415 = 2.002225$) so learners can test why $\sqrt{2}$ never ends. Note which learners need more practice with long division and plan a short support group.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We divided the numerator by the denominator to convert $1/2$, $1/4$, $1/8$, $1/3$, $1/6$, $1/7$, $1/15$ and $1/45$ into decimals. $1/2 = 0.5$, $1/4 = 0.25$, $1/8 = 0.125$ ended; $1/3 = 0.333\dots$, $1/6 = 0.1666\dots$, $1/7 = 0.142857142857\dots$, $1/15 = 0.0666\dots$, $1/45 = 0.02222\dots$ repeated. $\sqrt{2} = 1.41421356\dots$ and $\pi = 3.14159265\dots$ showed no end and no repeating block.
What did I learn?	Rational numbers can be written as a/b ($b \neq 0$) and have terminating or recurring decimals; whole numbers are rational ($n = n/1$) and Ksh $22.50 = 45/2$. Irrational numbers such as $\sqrt{2}$ and π have decimals that never end and never repeat. Rational and irrational numbers together make up the real numbers. Fractions and decimals are two ways of writing the same rational numbers.
How does this explain the phenomenon?	One shopping trip at the duka needs whole numbers, fractions and decimals because they are different ways of writing rational numbers for counting, sharing parts and pricing. They are all real numbers. Some numbers like $\sqrt{2}$ and π are irrational because their decimals never end or repeat, and we still need to find out why $\sqrt{2}$ goes on forever (DQB 5).

LESSON 4 (40 minutes): Why does $\sqrt{2}$ go on forever? Squaring checks on a square field

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence from a square plot of area 2 m^2 that $\sqrt{2}$ is irrational, and extend the unit model into a full real-number map.
Knowledge	b) classify real numbers as rational and irrational in different situations. Learners explain that the side of a square of area 2 m^2 is $\sqrt{2} = 1.41421356\dots$, that no ending decimal squares to exactly 2 ($1.414^2 = 1.999396$, $1.4142^2 = 1.99996164$, $1.415^2 = 2.002225$), and that rational and irrational numbers together make up the real numbers.
Skills	Analysing and interpreting data (squaring checks with a calculator); Constructing explanations; Engaging in argument from evidence; Developing and using models (number-system map). Digital Literacy: using the calculator. Communication and Collaboration: discussing with peers to categorise numbers as rational and irrational.
Attitudes	Unity: working in groups to categorise real numbers as rational and irrational. Responsibility: taking care of the calculators. Life Skills (self-esteem): participating confidently in the discussion on rational and irrational numbers. f) promote the use of real numbers in day-to-day activities.
Key Inquiry Question	How do we use real numbers in day-to-day activities? Supporting DQB question 5: Why does $\sqrt{2}$ go on forever?
Purpose in Storyline	Lesson 3 showed that rational numbers have terminating or recurring decimals while $\sqrt{2}$ and π never end or repeat. Lesson 2 taught that classification must be justified by evidence. This lesson gives the evidence for $\sqrt{2}$ using a square field, answers DQB question 5, and completes the map of real numbers (rational and irrational) in the unit model. It leads to Lesson 5 on reciprocals, which returns to unit price ($90 \div 6 = \text{Ksh } 15$) from Achieng's duka trip.
Safety Notes	Mark the 2 m^2 square plot on the school field in daylight, away from the road, with learners staying behind the marked lines. Use calculators carefully and keep them away from the edges of desks. If no field is available, mark the scale model with chalk or string on the classroom floor. Do not run while measuring.

B. LESSON OVERVIEW

Achieng's duka trip used whole numbers, fractions and decimals, and Ms. Wanjiku's class now asks whether every number in a shop or on a field can be written as a fraction. Learners imagine a square plot of area exactly 2 m^2 (a scale model on the school field). They predict whether its side can be measured exactly with a decimal. Using calculators, learners square 1.414, 1.4142 and 1.415 and find 1.999396, 1.99996164 and 2.002225. The side $\sqrt{2}$ lies between 1.4142 and 1.415, and no ending decimal squares to exactly 2. Groups (Lake Victoria, Kit Mikayi, Impala and Rift Valley) argue from evidence, explain why $\sqrt{2}$ goes on forever and never repeats, and then place the number types from the duka trip (3, 15, $\frac{1}{2}$, 22.50, $\sqrt{2}$, π) on a real-number map. They add this to the DQB and the unit model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Classifying	Learners sit in their groups (Lake Victoria, Kit Mikayi, Impala, Rift Valley). They recall Achieng at the	Say: 'Remember Achieng's trip. Today a farmer in Kisumu wants a square plot with an area of exactly	Predict-Observe-Explain (prediction stage) with a Think-Write-Share routine. Writing a	Look for: (1) every learner has written a prediction with a reason; (2) at least 3 of the 4 groups use

numbers: rational & irrational Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Classify numbers: rational & irrational Source: Khan Academy (English - US curriculum) Search ARES for similar readings	duka in Kisumu, where the numbers 3, 15, $\frac{1}{2}$ and Ksh 22.50 were all used. Ms. Wanjiku shows a drawing of a square plot of area exactly 2 m^2. Each learner writes in their notebook: 'The side of the square is ____ m. I think it is a number that (ends / repeats / never ends and never repeats) because ____.' Learners draw a quick sketch of the square and mark its side with a question mark. Groups share their predictions and record them on a chart.	2 square metres. How long is each side?' Draw the square on the board. Give 1 minute of silent writing and WAIT TIME of 10-15 seconds before anyone speaks. Cold-call 4 learners, one from each group, with: 'What is your prediction, and what is your reason?' Do not correct any answers. Write the predictions on the board under the headings 'Ends', 'Repeats' and 'Never ends or repeats'. Ask: 'Which prediction would we need evidence to test?' Link back to Lesson 3 by saying: 'Last lesson we saw that $\frac{1}{3}$ recurs but $\sqrt{2}$ never repeats. Today we find out why.'	prediction with a reason makes prior ideas visible, so learners later notice when the evidence supports or challenges them. Linking the prediction to the duka phenomenon keeps the focus on how number types are related.	words such as 'ends', 'repeats' or 'never ends'; (3) learners link the answer to decimals seen in Lesson 3 such as 0.5 or 0.333...; (4) record on a tally who predicts a terminating decimal (for example 1.4 or 1.41) as a baseline for later revision.
Observe Phase VIDEO: Approximating square roots Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Approximating square roots Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners go out to the school field, where the teacher has marked a square plot of area 2 m^2 (scale model) with string and pegs. Each group measures the side with a tape measure and finds that it is a little more than 1.4 m. Back in class, each group receives a calculator and fills in a table of squaring checks. They compute $1.414 \times 1.414 = 1.999396$ (just under 2), $1.4142 \times 1.4142 = 1.99996164$ (just under 2) and $1.415 \times 1.415 = 2.002225$ (over 2). They then use the $\sqrt{}$ key to display $\sqrt{2} = 1.41421356...$ and read off the digits. Each group records whether any decimal that ends gives exactly 2, and writes down which two numbers the side lies between (1.4142 and 1.415). They also try to find a repeating block in the digits and report that none appears.	Before going out, say: 'Calculators are shared tools. Hold them with both hands and keep them away from the edge of the desk.' (Responsibility value). On the field say: 'Measure the side of the square. Is the length exactly 1.4 m? Why is it hard to be exact?' Back in class, circulate and ask: 'What does 1.414×1.414 give? Is it above 2 or below 2?' Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 groups to read out one result each. Record the three results on the board in a table: $1.414 \rightarrow 1.999396$; $1.4142 \rightarrow 1.99996164$; $1.415 \rightarrow 2.002225$. Ask: 'What happens as we add one more decimal place? Does it ever hit exactly 2?' Prompt weaker groups: 'Look at the last digit of each answer. What does that tell you?' Remind learners that $\frac{1}{3}$ repeats as 0.333... and ask them to look at $\sqrt{2} = 1.41421356...$ for any repeating block.	Gather and Organise Data (analysing and interpreting data). A shared results table lets learners see a pattern across several trials: squares get closer to 2 but never reach it exactly with a decimal that ends. Linking the field measurement to the calculator result keeps it connected to a real situation.	Check that: (1) learners correctly compute and record all three squares (1.999396, 1.99996164, 2.002225); (2) at least 3 of 4 groups state that 1.414 and 1.4142 are 'just under 2' and 1.415 is 'over 2'; (3) learners say the side lies between 1.4142 and 1.415; (4) learners handle the calculators with care. Tick on a checklist as you circulate; support groups who misplace decimal points when multiplying.
Explain Phase	Each group prepares a claim-	Say: 'You are scientists reporting	Argue from Evidence using Claim-	Observable indicators: (1) CER

 VIDEO: Proof: $\sqrt{2}$ is irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Square Roots and Irrational Numbers (Flexbook) Source: CK-12  Search ARES for similar readings	evidence-reasoning card on the question 'Why does $\sqrt{2}$ go on forever?'. The claim: $\sqrt{2}$ is irrational. The evidence: the three squaring checks, the field measurement and the digits 1.41421356... that do not repeat. The reasoning: if $\sqrt{2}$ were rational, its decimal would end or repeat like $1/2 = 0.5$ or $1/7 = 0.142857...$, but no ending decimal squares to exactly 2 and no repeating block appears. Groups then hold a short class debate in which one group presents and another challenges ('Maybe the next digit makes it exact?'). Learners then sort numbers from Achieng's trip and earlier lessons into rational (3, 15, 65, $1/2$, 22.50, $1/3$, $1/7$, $1/45$) and irrational ($\sqrt{2}$, π) with reasons based on their decimals, and describe how together they form the real numbers.	to the farmer in Kisumu. Make a claim and back it with evidence from our table.' Provide sentence starters: 'Our claim is... Our evidence is... This shows that...'. Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to present, then ask the rest: 'Who can add evidence or challenge this?' (2 challengers). Press with: 'Someone says if we keep adding decimal places we will finally reach exactly 2. What would you say?' Guide learners to see that each ending decimal squares to a number that ends in a non-zero last digit, so it can never be exactly 2 (for example 1.4142^2 squared gives 1.99996164). Ask: 'Where does 22.50 go? Where does $1/2$ go?' Clarify that rational means written as a/b with $b \neq 0$, so terminating and recurring decimals are rational, and that rational and irrational numbers together are the real numbers. Do not give the proof; say that the formal proof comes in later years.	Evidence-Reasoning, plus a sorting task. Defending claims with data pushes learners from 'it just goes on' to 'no ending or repeating decimal can equal $\sqrt{2}$'. Sorting numbers from the duka trip applies the idea and links the classification to evidence, as in Lesson 2.	card has a claim, at least two pieces of evidence (the squaring checks, the non-repeating digits) and a reason linking to rational vs irrational; (2) learners correctly place 3, 15, $1/2$, 22.50 and $1/3$ as rational and $\sqrt{2}$ and π as irrational; (3) learners can say why 1.4142 is not exactly $\sqrt{2}$; (4) at least 3 learners ask or answer a challenge question politely (Unity, self-esteem). Common misconception to watch: 'a long decimal means irrational'. Redirect with $1/7 = 0.142857142857...$, which is long but repeats.
Driving Question Board (DQB) Creation  VIDEO: Intro to rational & irrational numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Square Roots and Irrational Numbers Practice Source: CK-12  Search ARES	Learners go to the DQB with sticky notes. They mark DQB question 5 ('Why does $\sqrt{2}$ go on forever?') with a green tick as partly answered and write a short answer on a note: 'No decimal that ends squares to exactly 2, and the digits never repeat.' They also look at questions 2 and 4 (What types of numbers exist? What's the difference between rational and irrational numbers?) and add new evidence notes. They post one new question each, for example: 'Are there other square roots like $\sqrt{3}$ that go on forever?' or 'What is a reciprocal, and what is the reciprocal of $\sqrt{2}$?'. Questions 6 and 7 are flagged as next.	Say: 'Go to the DQB. Which question can we now answer, and what is your evidence?' Give WAIT TIME of 10-15 seconds before learners post. Cold-call 3 learners to read their notes aloud and ask: 'What evidence supports your answer?' Keep the numbering of DQB questions 1-8 as supplied; do not renumber. Group new questions and point to questions 6 and 7: 'Next lesson we meet reciprocals, which link back to unit price, Ksh $90 \div 6 = \text{Ksh } 15$.' Celebrate quieter learners who contribute (self-esteem).	Driving Question Board as a progress tracker. Connecting notes to evidence shows what the class has learned and what is still unknown, so the next lesson has a clear purpose.	Check that: (1) every learner posts a note or a question; (2) the answer to question 5 cites the squaring evidence; (3) at least 2 new questions go beyond what was taught (for example about $\sqrt{3}$ or reciprocals); (4) learners use correct terms: rational, irrational, non-terminating, non-recurring.

for similar readings				
Model Building Phase  VIDEO: Classifying numbers: rational & irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers: rational & irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups revise the unit model, the cumulative number-system map. They add the real numbers as a large box that contains two parts: rational numbers (whole numbers such as 3, 15 and 65, fractions such as $\frac{1}{2}$, and terminating and recurring decimals such as 22.50, 0.5, 0.333... and 0.142857...) and irrational numbers ($\sqrt{2}$ and π , with decimals that never end or repeat). They add the square plot (area 2 m ² , side $\sqrt{2}$) as a labelled picture with the three squaring checks as evidence. They draw arrows from Achieng's duka numbers to the right place on the map and write one sentence: 'A number is rational if... and irrational if...'. Groups do a gallery walk and compare maps.	Say: 'Revise your model. Where do the whole numbers and fractions from the duka go? Where does $\sqrt{2}$ go? Show the evidence.' Give WAIT TIME of 10-15 seconds before groups begin drawing. Walk around and ask: 'Why is 22.50 inside rational numbers?' and 'Which evidence shows $\sqrt{2}$ is outside?'. For the gallery walk, cold-call 2 groups to explain one change they made and why. Ask: 'What does the map still not show?' and guide towards reciprocals for Lesson 5, as DQB questions 6 and 7 remain open. Do not teach reciprocals yet.	Developing and Using Models (revising a cumulative model). Updating the unit model with new evidence helps learners see how rational and irrational numbers fit together as real numbers, and connects the duka numbers to the field evidence.	Indicators: (1) the model shows real numbers containing rational and irrational numbers; (2) at least 5 duka numbers (3, 15, $\frac{1}{2}$, 22.50, Ksh 135) are placed correctly in rational numbers; (3) $\sqrt{2}$ and π appear under irrational with the reason 'decimal never ends or repeats'; (4) the square-field evidence appears as labelled support; (5) learners explain at least one change from their previous model. Collect exit slips: 'Write one sentence answering DQB question 5.'

D. TEACHER REFLECTION

Which groups moved from predicting that $\sqrt{2}$ ends or repeats to arguing that it does not, and what evidence convinced them? Did any learners confuse 'long decimal' with 'irrational'? How did the calculator work (digital literacy and care for the tools) go in each group? Did quieter learners take part in the debate (self-esteem, unity)? What would I change about the field activity or the squaring checks next time? Which DQB questions need to be revisited before Lesson 5 on reciprocals?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We squared 1.414, 1.4142 and 1.415 and got 1.999396, 1.99996164 and 2.002225. The first two are just under 2 and the last is over 2. The side of the square plot of area 2 m ² is between 1.4142 and 1.415. $\sqrt{2} = 1.41421356...$ and the digits do not end or repeat.
What did I learn?	No decimal that ends can square to exactly 2, and $\sqrt{2}$ has no repeating block, so $\sqrt{2}$ cannot be written as a fraction $\frac{a}{b}$: it is an irrational number. Rational numbers (whole numbers, fractions, terminating and recurring decimals) and irrational numbers (such as $\sqrt{2}$ and π) together make up the real numbers.
How does this explain the phenomenon?	$\sqrt{2}$ goes on forever because it is irrational: its decimal never ends and never repeats, unlike $\frac{1}{2} = 0.5$ or $\frac{1}{7} = 0.142857...$, which are rational. All the numbers on Achieng's duka trip (3, 15, $\frac{1}{2}$, Ksh 22.50) are rational real numbers, and the real-number map now also has a place for irrational numbers such as $\sqrt{2}$.

LESSON 5 (40 minutes): What is a Reciprocal? Sharing, Unit Price and Division

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners discover that the reciprocal of a number n is $1 \div n$ (written $1/n$), find reciprocals by division, and verify that $n \times 1/n = 1$, linking the idea to unit price at the duka in Kisumu.
Knowledge	c) determine the reciprocal of real numbers by division. Learners know that the reciprocal of n ($n \neq 0$) is $1/n = 1 \div n$, that $n \times 1/n = 1$, that zero has no reciprocal, and that unit price Ksh 15 = $90 \div 6 = 90 \times 1/6$.
Skills	Analysing and interpreting data (finding patterns in a table of reciprocals); Using mathematics and computational thinking (long division of 1 by n); Constructing explanations (why $n \times 1/n = 1$); Communication and Collaboration: the learner discusses with peers to identify the pattern in reciprocals.
Attitudes	Unity: learners work in groups (Lake Victoria, Kit Mikayi, Impala, Rift Valley) and share one agreed rule. Life Skills (self-esteem): every learner contributes a reciprocal and explains it aloud. Curiosity about why a simple pattern keeps working.
Key Inquiry Question	How do we use real numbers in day-to-day activities? Today's focus: What is a reciprocal? (DQB question 6, which also starts to answer question 7.)
Purpose in Storyline	In Lesson 1, Achieng met unit price: $90 \div 6 = \text{Ksh } 15$ for mandazi, a sharing situation. In Lesson 3, we converted fractions to decimals by dividing. Today these two ideas join: dividing 1 by a number gives its reciprocal, and dividing by 6 is the same as multiplying by $1/6$. This prepares Lesson 6 (tables and calculator) and Lesson 7 (speed, time, interest), where reciprocals answer the Driving Question.
Safety Notes	Classroom-based paper and pencil lesson. Keep walkways clear during group work. Learners who use calculators for checking should handle them with care (Responsibility). Be sensitive to learners who find division slow; praise method, not speed.

B. LESSON OVERVIEW

Achieng's mother asks, 'If 6 mandazi cost Ksh 90, what does one cost?' Mzee Odhiambo says the unit price is Ksh 15. Learners first predict what happens if we share Ksh 1 or 1 whole among n people. They then work in groups to find $1 \div n$ for 2, 4, 8, 6, 15, 45, $1/2$ and 2.5 using long division (building on Lesson 3's fraction-to-decimal division), and multiply each answer back by n to see that the product is always 1. They link the pattern to unit price: $90 \div 6 = 90 \times 1/6 = \text{Ksh } 15$. The lesson ends by adding DQB question 6 evidence and revising the unit model with a new branch: reciprocals.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dividing a whole number by a fraction with reciprocal	Ms. Wanjiku displays: 'Mama Atieno sells 6 mandazi for Ksh 90. Mzee Odhiambo says each costs Ksh 15.' Learners think quietly then write: (a) If 1 mandazi is one share of 6, what fraction of the	Display the duka scene and read it aloud. Ask: 'Achieng wants to know what unit price really means. If we share 1 whole into 6 equal parts, how big is one part?' WAIT TIME: 10-15 seconds in silence.	Predict-Observe-Explain (POE) with a Notice and Wonder prompt: committing to a prediction gives learners a stake in the evidence, exposes the prior idea that 'dividing by 6' and 'taking one sixth'	Look for: learners writing $1/6$ for one share; learners who predict $6 \times 1/6 = 1$ versus 6 or 7; learners who connect $90 \div 6$ to 'one sixth of 90'. Tally on the board how many predict that $6 \times 1/6$ is 1. Note

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 Search ARES for similar readings	pack is it? (b) Predict what $1 \div 6$ and $90 \times 1/6$ might equal. (c) Predict: if you multiply 6 by the share $1/6$, what will you get? Pairs share predictions, and all write in their notebooks as 'Prediction 5'.	Then say: 'Write your prediction before you talk.' Cold-call 4 learners from different groups (one each from Lake Victoria, Kit Mikayi, Impala and Rift Valley) to share predictions, and record all on the board without judging: 'I am not saying right or wrong yet; we will test these.' Ask one more learner: 'Who disagrees, and why?'	are different, and creates a reason to check later.	learners who give no prediction for a one-to-one check.
Observe Phase VIDEO: Dividing a whole number by a fraction with reciprocal Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Identification and Writing of Reciprocal Fractions Practice Source: CK-12 Search ARES for similar readings	In groups of four, learners complete the Reciprocal Evidence Table with long division of 1 by n: for each n they write $1 \div n$ and then check $n \times (1 \div n)$. Rows: $n = 2 \rightarrow 0.5$; $4 \rightarrow 0.25$; $8 \rightarrow 0.125$; $6 \rightarrow 0.1667$ (4 d.p., exact 0.1666...); $15 \rightarrow 0.0667$ (4 d.p.); $45 \rightarrow 0.02222$ (4 s.f.); $1/2 \rightarrow 1 \div 1/2 = 2$; $2.5 \rightarrow 0.4$. Each group handles all rows by splitting the work: each learner does two rows, then shares. Next they test the products: $2 \times 0.5 = 1$; $4 \times 0.25 = 1$; $8 \times 0.125 = 1$; $6 \times 1/6 = 1$; $45 \times 1/45 = 1$; $1/2 \times 2 = 1$; $2.5 \times 0.4 = 1$. For the recurring ones, they notice that the decimal is rounded, but the fraction product is exactly 1. Finally they test the duka link: $90 \div 6 = 15$ and $90 \times 1/6 = 90 \times 0.1666... = 15$.	Say: 'A reciprocal of n is what you get when you divide 1 by n. Today you will test whether it behaves in the same way each time.' Model one row on the board: $1 \div 4 = 0.25$, then $4 \times 0.25 = 1$. Circulate and ask: 'What do you notice about the product?' and 'Why does 6×0.1667 give 1.0002 instead of exactly 1?' WAIT TIME: 10-15 seconds after each question. Cold-call 3 groups to report one row each (one terminating, one recurring, one fraction/decimal such as $1/2$ or 2.5). Prompt struggling groups: 'Remember Lesson 3: $1/6$ was 0.1666..., a recurring decimal.' Ask a challenge question: 'What is $1 \div 0$? Try it. What happens?' (Zero has no reciprocal.)	Data table pattern-finding (Analysing and Interpreting Data): organising results in a table makes the constant product of 1 visible across different kinds of numbers (whole numbers, fractions, decimals), so the rule is built from evidence rather than told.	Observable indicators: correct long division set-up for $1 \div 6$ and $1 \div 45$ (adding zeros after the decimal point); correct rounding to 4 d.p. or 4 s.f.; written conclusion '$n \times 1/n = 1$' appears in at least 5 of the rows; learners say 'rounding makes the product slightly different from 1'. Use a tally sheet for each group: rows completed correctly out of 8.
Explain Phase VIDEO: Dividing whole numbers & fractions: t-shirts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Groups answer on a half-sheet: (1) In your own words, what is a reciprocal and how do you find it? (2) Use the table to explain why $n \times 1/n = 1$. (3) Explain Achieng's question using words and numbers: why is $90 \div 6$ the same as $90 \times 1/6$? (4) Which number has no reciprocal and why? Each learner then writes a Claim-	Say: 'Use the evidence in your table, not what you remember.' Give 3 minutes of writing time before talking. WAIT TIME: 10-15 seconds after asking, 'Why does the product always come to 1?' Cold-call 4 learners, each from a different group, to give one part of the explanation. Probe: 'Say more about sharing 1 among 6.' 'Who	Claim-Evidence-Reasoning (CER) with a revoicing talk move: requiring a claim backed by table data and a reason in words turns observation into explanation, and revoicing lets learners hear precise vocabulary.	Look for explanations that include 'divide 1 by n', '$n \times 1/n = 1$', and 'zero has no reciprocal'. Strong response: 'Unit price is dividing the total by the number of items, which is the same as multiplying by the reciprocal.' Check for the misconception where learners write that $1/n$ is 'n turned upside down' without any link to division

 Search ARES for similar videos  HTML: Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12  Search ARES for similar readings	Evidence-Reasoning statement: Claim: 'The reciprocal of n is $1 \div n$.' Evidence: table rows (for example $6 \rightarrow 0.1667$ and $6 \times 1/6 = 1$; $45 \rightarrow 0.02222$ and $45 \times 1/45 = 1$). Reasoning: 'Dividing 1 into n equal parts gives one part, and n of those parts make the whole 1 again.' Learners also answer: Is the reciprocal of $1/2$ equal to 2? and what is 0.5's reciprocal?	can say it in a different way?' Address the misconception that the reciprocal is the 'negative' or the 'opposite' of a number: 'Opposite means adding to get zero; reciprocal means multiplying to get 1.' Link back to the phenomenon: 'So when Mzee Odhiambo says Ksh 15, he took 90 and multiplied by $1/6$.'		by 1. Use a thumbs signal (up, sideways, down) for confidence after the explanation.
Driving Question Board (DQB) Creation  VIDEO: Dividing fractions and whole number word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Identification and Writing of Reciprocal Fractions Practice Source: CK-12  Search ARES for similar readings	Learners return to the DQB. They write on sticky notes what they now know about question 6 ('What is a reciprocal?') and begin evidence for question 7 ('How do reciprocals help us solve problems?'), for example: 'Unit price Ksh 15 = $90 \times 1/6$.' They place notes under DQB questions 6 and 7, mark question 6 as 'mostly answered', and write new questions, such as 'Can we find reciprocals faster with tables or a calculator?' and 'Where else do we divide a total into equal shares, such as speed or interest?' They also review questions 2-5 and 8 and note which have evidence so far (Lessons 1-4).	Say: 'Look at the DQB. Which question can we now answer, and what is the evidence?' WAIT TIME: 10-15 seconds. Cold-call 3 learners to place a note and read it aloud. Ask: 'Which new question should we investigate next lesson?' Point out that the next lesson uses mathematical tables and the calculator $1/x$ key, and that Lesson 7 uses reciprocals in speed, time and savings. Remind the learners to take care of the DQB and keep notes neat (Responsibility).	Driving Question Board with Gallery Walk: tracking questions publicly shows progress, helps learners see how each lesson answers a part of the Driving Question, and provides ownership of the next steps.	Observe whether learners place notes under the correct question numbers (6 and 7), whether notes contain evidence (numbers) and not just opinion, and whether at least 3 new, relevant questions are produced. Make a count of how many learners contribute a note.
Model Building Phase  VIDEO: Dividing fractions: $3/5 \div 1/2$ <i>Related topic: not an exact match for this lesson.</i>	Groups revise their unit model (the cumulative number-system map) from earlier lessons. Existing branches: whole numbers (odd, even, prime, composite), fractions and decimals, rational numbers, irrational numbers such as $\sqrt{2}$ and π, and real numbers. They add a new box: 'Reciprocal of $n = 1 \div n =$	Say: 'Add a reciprocal box to your model and connect it to the duka. Explain the connection in one sentence.' Circulate and ask: 'Where does the reciprocal sit in our map of real numbers?' WAIT TIME: 10-15 seconds. Cold-call 3 groups to show their models (Lake Victoria, Impala, Rift Valley). Ask	Model revision with a concept map: adding the new idea and drawing connecting arrows to the anchoring phenomenon forces learners to integrate reciprocal with number types and unit price, supporting a coherent story across the unit.	Check that each model includes: the definition $1/n = 1 \div n$, the product $n \times 1/n = 1$, at least 3 correct examples, the unit price link ($90 \times 1/6 = 15$), and the zero exception. Exit ticket: 'Write the reciprocal of 8 and prove it.' (0.125 and $8 \times 0.125 = 1$.) Record how many learners get it right.

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Decimal Division (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	$1/n$; $n \times 1/n = 1$; zero has none.' They draw an arrow from the duka: 'Unit price $90 \div 6 = 15 = 90 \times 1/6$.' Examples inside the box: $2 \rightarrow 0.5$, $4 \rightarrow 0.25$, $8 \rightarrow 0.125$, $6 \rightarrow 0.1667$, $15 \rightarrow 0.0667$, $45 \rightarrow 0.02222$, $1/2 \rightarrow 2$, $2.5 \rightarrow 0.4$. They also note that the reciprocal of a rational number is rational (for example $1/6 = 0.1666...$ is recurring), and then complete the Summary Table.	one group to state the rule in 15 words. Close: 'Next lesson we will see how mathematical tables and the calculator $1/x$ key give reciprocals, and how accurate they are.' Collect the Summary Table as an exit ticket.		
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D. TEACHER REFLECTION

Which learners confused reciprocal with the opposite (negative) or with simply 'flipping' without connecting to division? Did the table of 8 rows give enough evidence that $n \times 1/n = 1$, including for recurring decimals where the rounded product was not exactly 1? How well did learners link Ksh $15 = 90 \times 1/6$ to unit price? Which groups need more practice with long division of 1 by 6, 15 or 45 before Lesson 6, where we use mathematical tables and the calculator $1/x$ key? Did all voices (Lake Victoria, Kit Mikayi, Impala, Rift Valley groups) take part?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that dividing 1 by 2, 4, 8, 6, 15, 45, $1/2$ and 2.5 gives 0.5, 0.25, 0.125, 0.1667, 0.0667, 0.02222, 2 and 0.4, and that each number multiplied by its answer gives 1 (for example $6 \times 1/6 = 1$ and $45 \times 1/45 = 1$). Also $90 \div 6 = 15 = 90 \times 1/6$.
What did I learn?	The reciprocal of a number n is $1 \div n$, written $1/n$. It is found by division, and $n \times 1/n = 1$. Zero has no reciprocal. Recurring reciprocals such as $1/6$ are rounded to 4 d.p. or 4 s.f.
How does this explain the phenomenon?	Unit price, such as Ksh 15 for one mandazi, is found by dividing the total by the number of items, which is the same as multiplying by the reciprocal ($90 \times 1/6 = 15$). Sharing 1 whole into n equal parts gives $1/n$, and n of those parts make 1 again, so reciprocals turn division problems into multiplication.

LESSON 6 (40 minutes (or one double lesson of 80 minutes)): Reciprocals with Mathematical Tables and Calculators: How Accurate Is Each Tool?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find reciprocals of real numbers from mathematical tables and with the calculator $1/x$ key, compare the two tools with the division method from Lesson 5, and look after the tools they use.
Knowledge	d) determine the reciprocal of real numbers by use of mathematical tables and calculators. Mathematical tables give reciprocals to 4 significant figures, e.g. $1/7 \approx 0.1429$, while the calculator $1/x$ key shows more digits, $1/7 = 0.142857142...$ The recurring decimals from Lesson 3 must be rounded. Other values: $1/45 \approx 0.02222$, $1/6 \approx 0.1667$, $1/2.5 = 0.4$.
Skills	Reading values from reciprocal tables; using the calculator $1/x$ key; rounding to 4 d.p. or 4 s.f.; comparing results from tables, calculator and division; checking with $n \times 1/n \approx 1$; communicating and collaborating in groups; digital literacy.
Attitudes	Responsibility: the learner takes care of calculators and mathematical tables when working out the reciprocal of real numbers. Unity: learners work together in groups (Lake Victoria Group, Kit Mikayi Group, Impala Group, Rift Valley Group). Life Skills (self-esteem): the learner participates confidently in the group discussion.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	Lesson 5 showed that the reciprocal of n is $1 \div n$ and that the unit price Ksh 15 equals $90 \times 1/6$. Lesson 3 showed that recurring decimals never end. Learners now ask whether there is a quicker way than long division, and how exact the answer is. This lesson gives them two tools, tables and the calculator, so that Lesson 7 can apply reciprocals to speed, time and interest and answer the driving question. It supports DQB questions 6 and 7.
Safety Notes	Hand out tables and calculators carefully and collect them in the same way. Keep the tables dry and flat, and do not fold the pages or write on them. Press calculator keys gently and switch off the calculators when finished. Where calculators are limited, groups share one per group and take turns. Make sure there are clear walkways between desks.

B. LESSON OVERVIEW

Achieng, in Mzee Odhiambo's duka near Kibuye Market, was told that the unit price of 6 mandazi costing Ksh 90 is Ksh 15, which is $90 \times 1/6$. Mzee Odhiambo does not do long division each time he sells: he uses tools. Learners first predict how exact a tool's answer will be. They then gather evidence by finding $1/7$, $1/6$, $1/45$ and $1/2.5$ with three methods: division, mathematical tables and the calculator $1/x$ key. Next they compare the results, explain why tables give 0.1429 while the calculator gives 0.142857142..., and check with $n \times 1/n$. They add to the DQB (questions 6 and 7) and revise the unit model by adding a tools-and-accuracy branch to the reciprocal section. The responsibility value is built in through looking after the tables and calculators.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall Achieng at the	Say: "Remember Achieng's trip.	Predict-Observe-Explain	Look for: learners who connect $1/7$

 VIDEO: Meaning of the reciprocal Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Rounding to the Nearest 2 Decimal Places, or the Hundredths Place Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	duka: Mzee Odhiambo gave the unit price of mandazi at once ($90 \times 1/6 = \text{Ksh } 15$). On a prediction card each learner answers individually: (1) Can a tool give $1/7$ exactly as a decimal? (2) If tables and a calculator both give $1/7$, will the answers look the same? Why or why not? (3) Which would you trust more? They share with a partner and then with their group, using Lesson 3's idea that $1/7$ is a recurring decimal.	Mzee Odhiambo didn't spend ten minutes dividing. What tools might a shopkeeper use?" Wait 10 seconds. Write on the board: $1/7 = ?$ Say: "Write your prediction on your card first. This is your own thinking, there are no wrong predictions." Give 2 minutes. Cold-call 3 learners from different groups, asking: "Will the tool's answer be exact or rounded? Tell us why." After each answer, wait 10 seconds before the next. Say: "Who thinks something different?" Record the predictions in two columns on the board: 'exact' and 'rounded'. Keep the cards for later.	(prediction card). Committing to a prediction in writing means learners must revise their thinking when the evidence arrives. It links to Lesson 3 recurring decimals and Lesson 5 reciprocals by division.	to a recurring decimal from Lesson 3; learners who guess that tools give different numbers of digits; learners who say that a tool cannot show an endless decimal. Note on the clipboard the number of learners predicting 'exact' and 'rounded'. Misconception to catch: 'a calculator is always exact'.
Observe Phase  VIDEO: Meaning of the reciprocal Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Rounding to the Nearest 3 Decimal Places, or the Thousandths Place Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Handling the tools with care (responsibility), learners work in groups. Station 1, Division: they divide $1 \div 7$, $1 \div 6$ and $1 \div 45$ to 5 or 6 decimal places, as in Lesson 5. Station 2, Mathematical tables: they locate 7 in the reciprocal table and read 0.1429; they locate 4.5 and read 0.2222, and then use that to work out $1/45 = 0.2222 \div 10 = 0.02222$; they read 6 as 0.1667. Station 3, Calculator: they press 7 then the $1/x$ key (or $1 \div 7 =$) and record 0.142857142...; then $6 \rightarrow 0.1666666...$; $45 \rightarrow 0.0222222...$; $2.5 \rightarrow 0.4$. Learners record every result in a table with columns Number, Division, Tables, Calculator and Check ($n \times 1/n \approx 1$). Example check: $7 \times 0.1429 = 1.0003$, while $6 \times 0.1667 = 1.0002$. They also show that $1/2.5 = 0.4$ gives $2.5 \times 0.4 = 1$ exactly. As they work, they note how many digits each tool gave.	Before handing out materials say: "Tables and calculators belong to the whole class. Carry them with two hands, keep tables flat and dry, no folding pages, no writing on them." Model one example at the board: find $1/7$ in the table, reading across the row for 7 and down to the 'difference' columns where needed. Say: "Notice the table gives four digits only: 0.1429." Circulate. Ask: "Why does the table show 0.1429 and not 0.142857?" Wait 10-15 seconds. Prompt groups that finish early: "Does 6×0.1667 give exactly 1? How close?" Check that learners take turns on the tools (Unity). At the end, say: "Switch off and return the calculators and close the tables carefully." Collect results on a class chart.	Data table with multiple methods. Putting the three methods side by side makes the pattern of rounding visible. Carrying out the $n \times 1/n$ check ties back to Lesson 5's meaning of reciprocal.	Observable indicators: learner correctly reads 0.1429 for $1/7$ and 0.1667 for $1/6$ from the tables; learner writes $1/45 \approx 0.02222$ and explains the shift of the decimal point; learner presses $1/x$ correctly and records the digits; learner completes the $n \times 1/n$ check without prompting; learner handles tools carefully (a quick tick on the checklist for responsibility). Reteach: if a learner reads the wrong row, return to the model example with the finger-tracing method.
Explain Phase	Groups study the class chart and	Say: "Take 3 minutes with your	Claim-Evidence-Reasoning with	Listen and look for: learners who

 VIDEO: Rounding to the Nearest 2 Decimal Places, or the Hundredths Place <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Rounding to the Nearest 2 Decimal Places, or the Hundredths Place Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	write a short explanation to answer: (1) Why do the tables and the calculator give different digits for 1/7? (2) Which tool is more accurate, and when is the accuracy of tables good enough? (3) Why does $1/2.5 = 0.4$ look different from $1/7$? Learners use the ideas of recurring decimals (Lesson 3) and rounding to 4 d.p. or 4 s.f. They conclude that $1/7$ is a recurring decimal $0.142857142857\dots$, so a tool must round it, and $1/2.5$ terminates exactly at 0.4. Then they apply the tools to the phenomenon: unit price of the mandazi as $90 \times 1/6 \approx 90 \times 0.1667 = \text{Ksh } 15.003$, which rounds to Ksh 15. They decide what is sensible for money: rounding to Ksh 15. Each group shares their explanation in a whole-class talk and uses one sentence frame: 'The tables give ____ because ____, while the calculator gives ____.'	group. Use the chart as your evidence." Circulate and ask: "What does the dot-dot-dot in 0.142857142... tell you?" Wait 10-15 seconds. Use talk moves: "Can someone add on to that?" "Who can say it in different words?" Cold-call 4 learners, one from each group (Lake Victoria, Kit Mikayi, Impala, Rift Valley), to share. Ask: "Using 0.1667 for $1/6$, 90×0.1667 is 15.003. Is that a problem for Mzee Odhiambo's price?" Wait 10 seconds. Close with: "Tables are accurate to 4 significant figures, the calculator shows more. Both are rounded for a recurring decimal." Correct the idea that 'the calculator is exact' by pointing to the dots. Invite the learners to revisit their prediction cards and mark them right or revised.	sentence frames and talk moves. Groups state a claim about accuracy, support it with numbers from the chart, and link it to the reasoning about recurring decimals and rounding. This builds Communication and Collaboration and Critical Thinking.	state that $1/7$ is recurring so it must be rounded; learners who give evidence with specific digits (0.1429 vs 0.142857142...); learners who decide when rounding is acceptable (money to the nearest shilling); learners who revise their prediction card. Exit check on a slip: 'Write $1/45$ to 4 significant figures and say which tool you used.' Expected: 0.02222.
Driving Question Board (DQB) Creation  VIDEO: Rounding to the Nearest 3 Decimal Places, or the Thousandths Place <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Rounding to the Nearest 3 Decimal Places,	Learners return to the DQB. Groups look at questions 6 (What is a reciprocal?) and 7 (How do reciprocals help us solve problems?) and write on sticky notes what they have now learned: ways to find a reciprocal (division, tables, calculator $1/x$ key) and how accurate each is. They sort the DQB questions into answered, partly answered and not yet answered. They add new questions such as: 'Does every number have a reciprocal?' and 'How can reciprocals help with speed and Sacco savings?' Question numbers 1-8 are not changed; new questions are added after question 8.	Say: "Look at questions 6 and 7 on our board. What can we now say about them with evidence?" Wait 10-15 seconds. Cold-call 3 learners to read a note and say which evidence supports it. Ask: "What are we still wondering?" Add new questions without renumbering 1-8. Point to question 8, 'Where do we use real numbers in daily life?', and say: "Next lesson we will use reciprocals for a bus journey from Nairobi to Mombasa and for Sacco savings." Thank the groups for careful use of the tools.	Driving Question Board as a living record. Learners tie each piece of new evidence to a question, so they see their understanding growing and what to investigate next.	Check that sticky notes cite evidence (e.g. 'tables give 0.1429 for $1/7$') and not just opinions. Look for sensible new questions leading to application. Note learners who can say which questions are answered and which are not. Look for participation from quieter learners (self-esteem).

or the Thousandths Place Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings				
Model Building Phase  VIDEO: Dividing whole numbers & fractions: t-shirts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Rounding to the Nearest 2 Decimal Places, or the Hundredths Place Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners update the unit model, the cumulative number-system map. In the reciprocal section they add: 'Reciprocal of $n = 1/n = 1 \div n$; $n \times 1/n = 1$ (zero has no reciprocal)', and three methods: division, mathematical tables (4 s.f.), calculator $1/x$ key (more digits). They add worked examples: $1/7 \approx 0.1429$ (tables) and $0.142857142...$ (calculator); $1/6 \approx 0.1667$; $1/45 \approx 0.02222$; $1/2.5 = 0.4$. They add an arrow from 'recurring decimals' (Lesson 3) to 'rounding' and a link to the duka: unit price Ksh 15 = $90 \times 1/6$. They also add a responsibility note: 'Care for the tables and calculators'. Individually, each learner writes one sentence that explains what is new in their model.	Say: "Open your unit model. Add what you now know about finding reciprocals and how exact each method is. Show the duka link." Allow 4 minutes. Circulate and ask: "Where in your model does the recurring decimal from Lesson 3 appear?" Wait 10 seconds. Cold-call 3 learners to share their additions. Ask: "What does your model still not explain?" Point out that Lesson 7 will use reciprocals in speed, time, rate and interest. Close by asking learners to pack away the tables and calculators neatly.	Cumulative model revision. Learners add new evidence and connections to their unit model, so the idea of the reciprocal is built up over several lessons from the single duka phenomenon.	Look for: models showing all three methods; correct examples (0.1429, 0.1667, 0.02222, 0.4); an arrow connecting recurring decimals to rounding; the unit price link $90 \times 1/6 = \text{Ksh } 15$; a note about care for tools. Collect the one-sentence reflections to check understanding, and use them to plan Lesson 7.

D. TEACHER REFLECTION

After the lesson ask yourself: Did learners see that a recurring decimal such as $1/7$ must be rounded and that a calculator shows more digits but is still not exact? Which learners read the tables wrongly, and was it because of the row or the decimal place? Were the tables and calculators handled with care, and did every learner have a turn? Did the quieter learners speak in their groups? Which learners could link $1/6$ and $1/45$ to the unit price at the duka? What will you change for Lesson 7, for example, more practice with the table layout or a quick recap of $1/x$ and $n \times 1/n = 1$?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We found $1/7$, $1/6$, $1/45$ and $1/2.5$ using division, mathematical tables and the calculator $1/x$ key. Tables gave $1/7 \approx 0.1429$, $1/6 \approx 0.1667$ and $1/45 \approx 0.02222$. The calculator gave $1/7 = 0.142857142...$ and $1/2.5 = 0.4$.
What did I learn?	Reciprocals can be found by division, from mathematical tables (4 significant figures) and with the calculator $1/x$ key (more digits). Recurring decimals like $1/7$ must be rounded, while $1/2.5 = 0.4$ ends exactly. We check with $n \times 1/n \approx 1$, and we look after the tables and calculators.
How does this explain the phenomenon?	Mzee Odhiambo can quickly find the unit price of the mandazi, $90 \times 1/6 = \text{Ksh } 15$, because the reciprocal of 6 is 0.1667 (rounded), and the tools give results that are accurate enough for money. The tables give fewer digits than the calculator because $1/6$ and $1/7$ are recurring decimals that never end, so every tool has to round them.

LESSON 7 (80 minutes (2 x 40-minute periods)): Reciprocals in Speed, Time, Rate and Interest: Answering the Driving Question

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply reciprocals to speed-time-rate and Sacco simple-interest problems, then synthesise the whole unit to write a Final Explanation answering the Driving Question: How do we use real numbers in everyday life, and why are reciprocals important?
Knowledge	By the end of the lesson the learner should be able to: e) apply reciprocal of real numbers in mathematical computations (time = distance x 1/speed; years = interest wanted x 1/interest per year); f) promote the use of real numbers in day-to-day activities. Learners also draw together the unit: number types (L1), classification of whole numbers (L2), rational and irrational numbers (L3, L4), reciprocal by division (L5) and tables and calculator (L6).
Skills	Critical Thinking and Problem Solving: the learner uses the reciprocal of real numbers to work out mathematical tasks. Communication and Collaboration: the learner discusses with peers to categorise numbers and defend claims. Digital Literacy: the learner works out the reciprocal of real numbers from calculators. NGSS-aligned practices: Engaging in Argument from Evidence, Developing and Using Models, Constructing Explanations, Using Mathematics and Computational Thinking.
Attitudes	Unity: learners work in groups (Lake Victoria Group, Kit Mikayi Group, Impala Group, Rift Valley Group) to categorise real numbers and build the final unit model. Responsibility: learners take care of calculators and mathematical tables. Life Skills (self-esteem): every learner takes part in explaining rational and irrational numbers. Appreciation of real numbers in day-to-day activities such as travel and saving.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This is the closing lesson. Achieng's duka trip in Kisumu ($3 \times 45 = \text{Ksh } 135$, change Ksh 65, $90 / 6 = \text{Ksh } 15$ unit price, 1/2 kg sugar at Ksh 80, tea sachet Ksh 22.50) raised the puzzle of why many types of numbers are needed. Lessons 1-6 gave the evidence. Here learners show reciprocals working beyond the duka (Nairobi to Mombasa journey, Wanjiru's Sacco savings), close the loop on 'unit price' as multiplying by a reciprocal, review all eight DQB questions, compare the Lesson 1 model with a final unit model, and write the Final Explanation.
Safety Notes	No hazardous materials. Learners handle calculators and mathematical tables carefully and return them after use. Keep walkways clear during gallery walks. Calculators are shared fairly within groups. No physical field movement is required.

B. LESSON OVERVIEW

Learners begin by predicting how long a bus takes from Nairobi to Mombasa (480 km) at 60 km/h and at 80 km/h, and how long Wanjiru must save in a Sacco (Ksh 10 000 at 8% simple interest per year) to earn Ksh 2 400. In the Observe phase, groups work out time = distance x 1/speed ($480 \times 1/60 = 8$ h; $480 \times 1/80 = 6$ h), the rate of 1/60 hour per km (1 minute per km), and the Sacco interest (Ksh 800 per year, Ksh 400 for 6 months, $2\,400 \times 1/800 = 3$ years), checking each with a calculator 1/x key or division. In the Explain phase they connect these to the duka unit price ($90 \times 1/6 = \text{Ksh } 15$) and sort every number met in the unit as whole, rational or irrational. They then review the DQB, answering questions 1-8 with evidence. Finally they revise the Lesson 1 model into a final unit model and write a Final Explanation answering the Driving Question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to interest Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12  Search ARES for similar readings	Learners individually write a prediction in their notebooks (5 minutes) for two situations. (a) A bus travels from Nairobi to Mombasa, 480 km: how many hours at 60 km/h, and at 80 km/h? (b) Wanjiru in Nairobi saves Ksh 10 000 in a Sacco at 8% simple interest per year: how much interest does she earn in 1 year, and how long to earn Ksh 2 400? They also write one sentence: 'I think reciprocals help here because ...' or 'I do not think reciprocals help because ...'. They then share with a partner and note where they disagree. Learners link back to Achieng's duka trip, where Mzee Odhiambo said the unit price of 6 mandazi at Ksh 90 is Ksh 15.	Display the two situations on the board. Say: 'Achieng's trip showed us many kinds of numbers. Today we find out whether the same ideas help on a bus to Mombasa and in a Sacco in Nairobi. Write your honest prediction first; wrong predictions are welcome.' Give WAIT TIME of 15 seconds after reading each question before learners write. Circulate and note 3 different predictions without correcting them. Cold-call 4 learners (one from each group: Lake Victoria, Kit Mikayi, Impala, Rift Valley) to share a prediction and the reason. Record all predictions on the board under 'Our predictions'. Ask: 'Who predicted something different from their partner? What made you disagree?' WAIT TIME 10 seconds before taking answers.	Predict-and-Commit (Developing and Using Models): learners commit to a prediction and a reason in writing before seeing evidence. This surfaces prior ideas about division and reciprocals from Lessons 5 and 6, so that later calculations can confirm or challenge them and learners see their own thinking change.	Observe whether learners (1) write a numerical prediction rather than leaving the space blank; (2) mention division, 'distance divided by speed' or 'multiplying by 1/6' in their reasoning; (3) can say what 'unit price' meant in the duka. Tally on a class list how many predicted 8 h for 60 km/h. Note learners who have no reasoning so that you can prompt them with the question 'what did we do with 90 and 6 at the duka?' during the Observe phase.
Observe Phase  VIDEO: Finding average speed or rate Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12  Search ARES for similar readings	Groups of 4 (Lake Victoria Group, Kit Mikayi Group, Impala Group, Rift Valley Group) complete a three-station evidence task (about 25 minutes). Station A, Journey: use distance 480 km. Work out time = $480 / 60 = 8$ hours and $480 / 80 = 6$ hours by division. Then rewrite as time = distance $\times$ 1/speed: $480 \times 1/60 = 8$ and $480 \times 1/80 = 6$. Find the reciprocal of 60 and 80 on the calculator 1/x key and in the mathematical tables (4 significant figures), compare and record. Rate: show that 1/60 hour per km is 1 minute per km. Station B, Sacco: Wanjiru's Ksh 10 000 at 8% per year. Use $I = PRT/100$: 1 year = $10\,000 \times 8/100 = \text{Ksh } 800$; 6 months ($T = 1/2$) = Ksh 400. To earn Ksh 2 400, time = $2\,400 \times 1/800 = 3$ years. Check: $800 \times 3 =$	Say: 'At each station, do the division first, then do it again by multiplying by the reciprocal. Do your two answers match?' Circulate with a checklist. Prompt: 'Which number in $480 \times 1/60$ is the reciprocal? What does 1/60 tell you about one kilometre?' Give WAIT TIME 10 seconds before a group answers. If a group uses the wrong time (for example 6 months as 6), ask: 'How many years is 6 months?' If a group says the reciprocal of 7 is exactly 0.1429, remind them of Lesson 6 (tables give 4 digits, the calculator gives more) and ask which is exact. Cold-call 2 learners per station (6 in total) to state the answer and the check. Remind learners to handle calculators and tables with care and to share them	Data Table Comparison (Using Mathematics and Computational Thinking): learners place the division method beside the reciprocal method for each situation, so they see that both give the same answer (8 h, 6 h, 3 years, Ksh 15). Seeing identical results from two methods builds the idea that a reciprocal turns division into multiplication.	Check the group tables for: $480 \times 1/60 = 8$ and $480 \times 1/80 = 6$; $10\,000 \times 8/100 = 800$; 400 for 6 months; $2\,400 \times 1/800 = 3$ years; the check $800 \times 3 = 2\,400$; and $90 \times 1/6 = 15$. Indicators of success: learners state 'x 1/speed' without prompting; they use the product $n \times 1/n = 1$ to check; they correctly convert 6 months to 1/2 year. Record learners who need support with converting time units, and pair them with a confident partner.

	2 400. Station C, Duka link: re-do $90 \div 6 = 15$ as $90 \times \frac{1}{6} = 15$, and check $6 \times \frac{1}{6} = 1$. Each learner records results in a table with columns: situation, division method, reciprocal method, check. Learners compare their predictions with the evidence.	(Responsibility). Stop at 25 minutes and ask each group to put one result on the board.		
Explain Phase  VIDEO: Solving for time Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Simple Interest Equation to Find Interest Rate Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners explain the evidence in three parts (about 20 minutes). Part 1: In groups, write a claim-evidence-reasoning (CER) statement: 'A reciprocal helps solve speed, time and interest problems because ...' using numbers such as $480 \times \frac{1}{80} = 6$ and $2\,400 \times \frac{1}{800} = 3$. Part 2: Sort the numbers from the unit into a number-type map. Whole numbers: 3, 6, 45, 90, 135, 200, 65, 15 (classified as odd, even, prime or composite from Lesson 2, for example 3 prime and odd, 6 even composite, 15 odd composite). Fraction: $\frac{1}{2}$ kg of sugar. Decimal: Ksh 22.50 = $\frac{45}{2}$. Reciprocal results such as $\frac{1}{7} = 0.142857...$ (recurring, rational). Irrational: $\sqrt{2} = 1.41421356...$ and $\pi = 3.14159265...$ Together these are real numbers. Part 3: Return to the Driving Question and the duka puzzle: explain why one shopping trip needs whole numbers, fractions and decimals, and what unit price really means (total cost / number of items, which is the same as multiplying by the reciprocal, $90 \times \frac{1}{6} = \text{Ksh } 15$). Learners hold a gallery walk of group explanations and give one 'star' and one 'wonder' on sticky notes.	Say: 'Use a claim, evidence from today's numbers, and a reason that links to what we learned about reciprocals.' Give WAIT TIME 15 seconds for thinking before groups write. Prompts: 'Why does a reciprocal make a division problem a multiplication problem? What did we check with $n \times \frac{1}{n} = 1$?' 'Which numbers on the map are rational, and how do you know? Which are irrational, and what evidence did the square field give us?' (Lesson 4 showed that no ending decimal squares to exactly 2.) Cold-call 5 learners to defend a placement on the map. Challenge: 'Achieng says 22.50 is not a whole number, so it is not a real number. Do you agree?' WAIT TIME 10 seconds, then take 3 responses. Do not re-teach earlier lessons; point to the class charts from Lessons 2 to 6 for evidence. Lead the gallery walk and ensure everyone speaks (self-esteem).	Claim-Evidence-Reasoning with Gallery Walk (Constructing Explanations; Engaging in Argument from Evidence): learners link a claim to specific numerical evidence and a reason, then critique peer explanations. Placing every number on the type map shows that whole, rational and irrational numbers are all real numbers, and applying the reciprocal to the unit price closes the loop on Achieng's puzzle.	Use a 3-point rubric on the CER statements: (1) a clear claim; (2) at least two correct numerical pieces of evidence (such as $480 \times \frac{1}{60} = 8$ and $90 \times \frac{1}{6} = 15$); (3) a reason that connects the reciprocal to turning division into multiplication. Listen for correct use of 'rational', 'irrational', 'terminating', 'recurring' and 'real numbers'. Look for misconceptions such as 'a recurring decimal is irrational' or 'zero has a reciprocal'. Note who is not contributing in groups and prompt them directly.
Driving Question Board (DQB)	Learners complete the Driving Question Board (about 10 minutes). Each group takes two of	Say: 'Look back at the DQB we started in Lesson 1. Which questions can we now answer with	DQB Review and Reflection (Asking Questions and Defining Problems): returning to the	Check each card for (1) a correct answer, (2) a reference to specific evidence (for example 'Lesson 4:

Creation  VIDEO: Finding average speed or rate Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mixtures and combined rates word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	the eight DQB questions and writes an evidence-based answer on a card, then sticks it beside the question. Question 1 (KICD key inquiry): How do we use real numbers in day-to-day activities? Question 2: What types of numbers exist? Question 3: How do we classify whole numbers? Question 4: What's the difference between rational and irrational numbers? Question 5: Why does $\sqrt{2}$ go on forever? Question 6: What is a reciprocal? Question 7: How do reciprocals help us solve problems? Question 8: Where do we use real numbers in daily life? Learners mark each question 'answered', 'partly answered' or 'new question', and write at least one new question, for example 'Do all reciprocals of whole numbers recur?' or 'How would a bank compound interest?' They read the cards from other groups and add a tick or a correction.	evidence, not just opinion?' Assign questions: Lake Victoria Group 1 and 8; Kit Mikayi Group 2 and 3; Impala Group 4 and 5; Rift Valley Group 6 and 7. Give WAIT TIME 15 seconds for groups to find the evidence before writing. Walk around and ask: 'Which lesson gave you this evidence? What number proves it?' Cold-call 4 learners to read a card aloud, one per group. Ask the class to vote with thumbs: fully answered, partly or still wondering. Keep unanswered questions visible for future lessons such as Indices and Logarithms, and say that new questions are a sign of good scientists and mathematicians.	learners' own questions shows how evidence from Lessons 1 to 7 answered them and generates new questions. This gives learners ownership of the unit and reveals what was learned.	$1.4142 \times 1.4142 = 1.99996164$, still under 2, so $\sqrt{2}$ does not end'), and (3) correct vocabulary. Indicators: learners cite evidence rather than 'because the teacher said'; at least one new, thoughtful question per group; correct corrections of peer cards. Record the questions marked 'partly answered' to plan revision.
Model Building Phase  VIDEO: Introduction to interest Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Related Rates of Car Speeds (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners finalise the unit model (about 15 minutes). Individually, they take out the Lesson 1 initial model (types of numbers from the duka: whole, fraction, decimal) and, in groups, draw a final unit model on chart paper showing: (1) real numbers as the whole set; (2) rational numbers (whole numbers, fractions, terminating and recurring decimals) and irrational numbers ($\sqrt{2}$, π); (3) whole numbers classified as odd, even, prime, composite; (4) the reciprocal $1/n$ with $n \times 1/n = 1$ and the methods of division, mathematical tables and calculator $1/x$ key; (5) applications: duka unit price $90 \times 1/6 = \text{Ksh } 15$, journey $480 \times 1/80 = 6$ h, Sacco $2\,400 \times 1/800 = 3$ years. Learners compare it with	Say: 'Your first model only listed kinds of numbers. What can your new model explain that the first one could not?' Give WAIT TIME 15 seconds before groups begin drawing. Remind them to include arrows showing relationships, not just lists. Cold-call 4 learners to name one change between the first and final model. Ask: 'Where does unit price fit on your model? Where does $\sqrt{2}$ fit, and why is it not a fraction?' WAIT TIME 10 seconds. Display the Final Explanation success criteria: (a) names whole numbers, fractions and decimals from the duka and says they are real numbers; (b) explains rational vs irrational with an example; (c) defines reciprocal and gives $n \times 1/n = 1$; (d) uses a	Model Revision and Then-and-Now Comparison (Developing and Using Models): learners compare the Lesson 1 model with the final unit model so they can see how evidence from every lesson added relationships (classification, rational vs irrational, reciprocals, applications). Writing the Final Explanation consolidates the learning into a coherent argument that answers the Driving Question.	Score the final unit model and Final Explanation against the criteria. Look for: all number types placed correctly on the map (whole numbers, $1/2$, 22.50, $\sqrt{2}$, π); the link between fractions and decimals as rational numbers; the reciprocal relationship $n \times 1/n = 1$ and zero having no reciprocal; at least one correct application ($480 \times 1/60 = 8$, or $2\,400 \times 1/800 = 3$, or $90 \times 1/6 = 15$). Success indicators: accurate vocabulary, evidence-based reasoning and a clear answer to 'why are reciprocals important?' Use the results to identify learners needing follow-up.

	the Lesson 1 model using a two-column 'Then and Now' chart and underline what changed. Finally each learner writes an individual Final Explanation (6 to 8 sentences) answering the Driving Question: How do we use real numbers in everyday life, and why are reciprocals important? It must use Achieng's duka trip, at least one number from the journey or Sacco, and explain why fractions and decimals are two ways of writing rational numbers.	numerical example from the unit; (e) says why reciprocals are important. Collect explanations and give feedback in the next lesson. Close by asking 3 learners to read a sentence aloud. End with: 'Next time you go to a duka, a matatu stage or a Sacco, which real numbers will you notice?'		
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D. TEACHER REFLECTION
After the lesson, reflect: Did learners predict that reciprocals would help, and did their view change after the calculations? Which learners correctly converted 6 months to $\frac{1}{2}$ year and used $2\,400 \times \frac{1}{800} = 3$ years without help? Did learners confuse recurring decimals (such as $\frac{1}{7}$) with irrational numbers (sqrt 2, pi)? Could they place unit price Ksh 15 as $90 \times \frac{1}{6}$ on the map? Which DQB questions remain 'partly answered'? Was WAIT TIME of 10 to 15 seconds enough to include quieter learners, and did all groups share calculators and tables responsibly? Note adjustments for the next sub-strand, such as more practice on time units, and plan to return the Final Explanations with feedback.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that a journey of 480 km takes 8 hours at 60 km/h and 6 hours at 80 km/h ($480 \times \frac{1}{60} = 8$, $480 \times \frac{1}{80} = 6$), that Wanjiru's Ksh 10 000 at 8% earns Ksh 800 a year and Ksh 400 in 6 months, and that Ksh 2 400 interest takes $2\,400 \times \frac{1}{800} = 3$ years. We also saw that the duka unit price $90 / 6 = \text{Ksh } 15$ is the same as $90 \times \frac{1}{6}$.
What did I learn?	We learned that real numbers include whole numbers, fractions and decimals (rational numbers) and numbers like sqrt 2 and pi (irrational numbers); that whole numbers are odd, even, prime or composite; that the reciprocal of n is $\frac{1}{n}$ and $n \times \frac{1}{n} = 1$; and that reciprocals can be found by division, mathematical tables or the calculator $1/x$ key.
How does this explain the phenomenon?	We explain that Achieng's one shopping trip needs whole numbers for counting, fractions for parts and decimals for prices because they are different ways of writing real numbers, and that reciprocals are important because they turn division into multiplication in unit price, speed, time, rate and interest problems, so real numbers and reciprocals together describe everyday life.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.